

fans that return part of the air to the distribution system. The remaining exhaust air passes to an outflow valve, which directs the air overboard. The cabin ventilation system is designed to deliver air uniformly along the length of the cabin.

Pneumatic System

The pneumatic system, or **engine bleed air system**, extracts a small amount of the gas turbine engine compressor air to ventilate and pressurize the aircraft compartments. A schematic of a typical system is shown in Figure 10. During climb and cruise, bleed air is usually taken from the mid-stage engine bleed port for minimum-horsepower extraction (bleed penalty). During idle descent it is taken from the high-stage engine bleed port, where maximum available pressure is required to maintain cabin pressure and ventilation. The auxiliary power unit (APU) is also capable of providing the pneumatic system with compressed air on the ground and in flight. Bleed air is pressure-controlled to meet the requirements of the system using it, and it is usually cooled to limit bleed manifold temperatures to meet fuel safety requirements. In fan jets, engine fan air is extracted for use as a heat sink for bleed air using an air-to-air heat exchanger called a precooler; for turboprop engines, ram air is used, which usually requires an ejector or fan for static operation. Other components include bleed-shutoff and modulating valves, a fan-air-modulating valve, sensors, controllers, and ozone converters. The pneumatic system is also used intermittently for airfoil and engine cowl anti-icing, engine start, and several other pneumatic functions.

Each engine has an identical bleed air system for redundancy and to equalize the compressor air bled from the engines. The equipment is sized to provide the necessary temperature and airflow for airfoil and cowl anti-icing, or cabin pressurization and air

conditioning with one system or engine inoperative. The bleed air used for airfoil anti-icing is controlled by valves feeding piccolo tubes extending along the wing leading edge. Similar arrangements may be used for anti-icing the engine cowl and tail section.

Air Conditioning

Air-cycle refrigeration is the predominant means of air conditioning for commercial and military aircraft. The reverse-Brayton cycle or Brayton refrigeration cycle is used, as opposed to the Brayton power cycle that is used in gas turbine engines. The difference between the two cycles is that, in the power cycle, fuel in a combustion chamber adds heat, and in the refrigeration cycle, a ram-air heat exchanger removes heat. The familiar Rankine vapor cycle, which is used in building and automotive air conditioning and in domestic and commercial refrigeration, is used for military aircraft as well as galley cooling on larger commercial transports.

In an air cycle, compression of the ambient air by the gas turbine engine compressor provides the power input. The heat of compression is removed in a heat exchanger using ambient air as the heat sink. This cooled air is refrigerated by expansion across a turbine powered by the compressed bleed air. The turbine energy resulting from the isentropic expansion is absorbed by a second rotor, which is either a ram air fan, bleed air compressor, or both. This assembly is called an **air cycle machine (ACM)**.

The most common types of air-conditioning cycles for commercial transport aircraft are shown in Figure 11. All equipment in common use on commercial and military aircraft is open loop, although many commercial aircraft systems include various means of recirculating cabin air to minimize engine bleed air use without sacrificing cabin comfort. The basic differences between the systems are the type of air cycle machine used and its means of water separation. Hybrid ACM/vapor cycle systems are discussed in Chapter 27 of the 2018 *ASHRAE Handbook—Refrigeration*.

The most common of these air cycle machines in use are the bootstrap ACM consisting of a turbine and compressor; the three-wheel ACM consisting of a turbine, compressor, and fan; and the four-wheel ACM consisting of two turbines, a compressor, and a fan. The bootstrap ACM is most commonly used for military applications, although many older commercial aircraft models use the bootstrap cycle. The three-wheel ACM (simple bootstrap cycle) is used on most of the newer commercial aircraft, including commuter aircraft and business aircraft. The four-wheel ACM (condensing cycle) was first applied in 777 aircraft.

The compartment supply temperature may be controlled by mixing ram-cooled bleed air with the refrigerated air to satisfy the range of heating and cooling. Other more sophisticated means of temperature control are often used; these include ram air modulation, various bypass schemes in the air-conditioning pack, and downstream controls that add heat for individual zone temperature control.

The bleed airflow is controlled by a valve at the inlet of the air-conditioning pack. The flow control valve regulates flow to the cabin for ventilation and repressurization during descent. Most aircraft use two or three air cycle packs operating in parallel to compensate for failures during flight and to allow the aircraft to be dispatched with certain failures. However, many business and commuter aircraft use a single pack. High-altitude aircraft that have a single pack also have emergency pressurization equipment that uses ram-cooled bleed air.

If the engine ingests water, or if the air cycle drops significantly below the dew point, some water separation devices are installed to avoid water spray in the cabin. Low- or high-pressure water separation may be used. A **low-pressure water separator**, located downstream from the cooling turbine, has a cloth lining that coalesces fine water particles entrained in the turbine discharge air into droplets. The droplets are collected, drained, and sprayed into the ram airstream using a bleed-air-powered ejector; this process

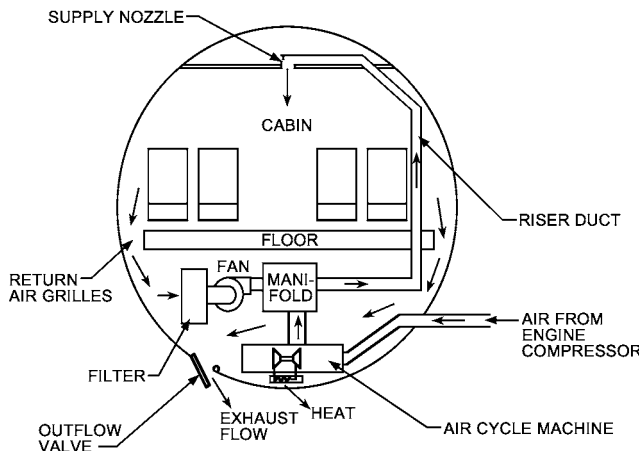


Fig. 9 Cabin Airflow Path

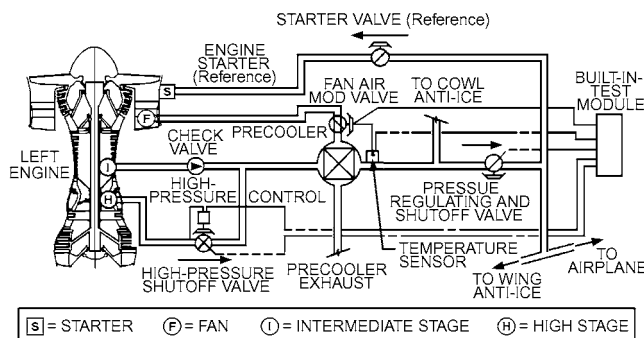


Fig. 10 Engine/APU Bleed System

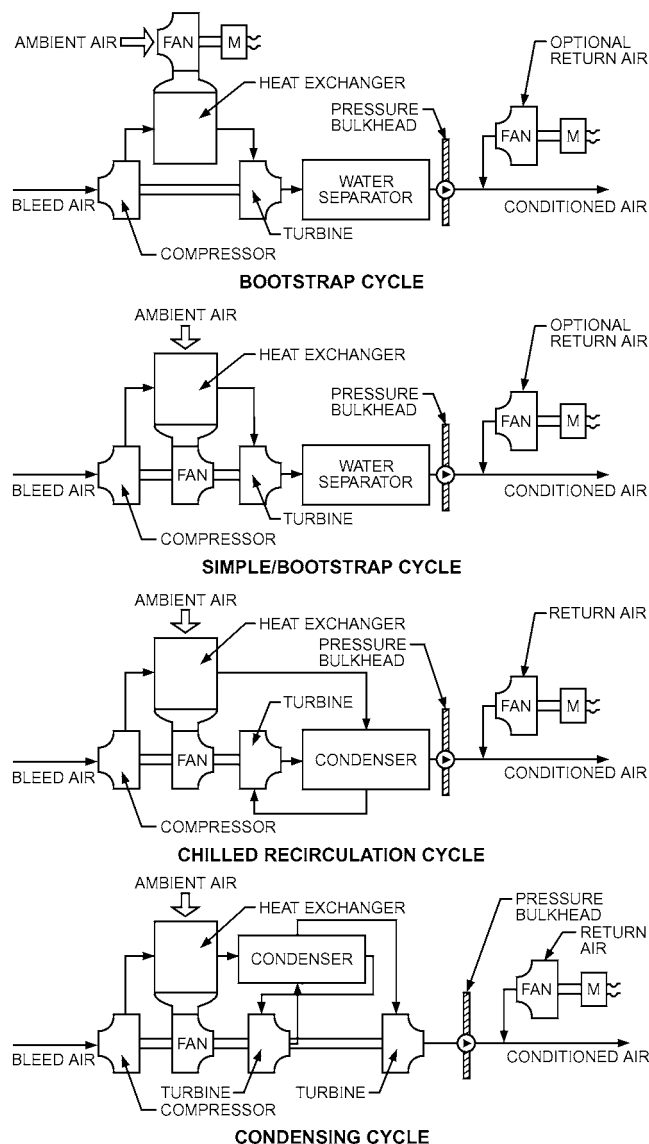


Fig. 11 Some Aircraft Refrigeration Cycles

increases pack cooling capacity by depressing the ram air heat sink temperature.

The **high-pressure water separator** condenses and removes moisture at high pressure upstream of the cooling turbine. A heat exchanger uses turbine discharge air to cool the high-pressure air sufficiently to condense most of the moisture present in the bleed air supply. The moisture is collected and sprayed into the ram airstream.

In the condensing cycle one turbine removes the high-pressure water and the second turbine does the final expansion to subfreezing temperature air that is to be mixed with filtered, recirculated cabin air. Separating these functions recovers the heat of condensation, which results in a higher cycle efficiency. It also eliminates condenser freezing problems because the condensing heat exchanger is operated above freezing conditions.

The air-conditioning packs are located in unpressurized areas of the aircraft to minimize structural requirements of the ram air circuit that provides the necessary heat sink for the air-conditioning cycle. This location also provides protection against cabin depressurization in the event of a bleed or ram air duct rupture. The most common areas for the air-conditioning packs are the underwing/wheel

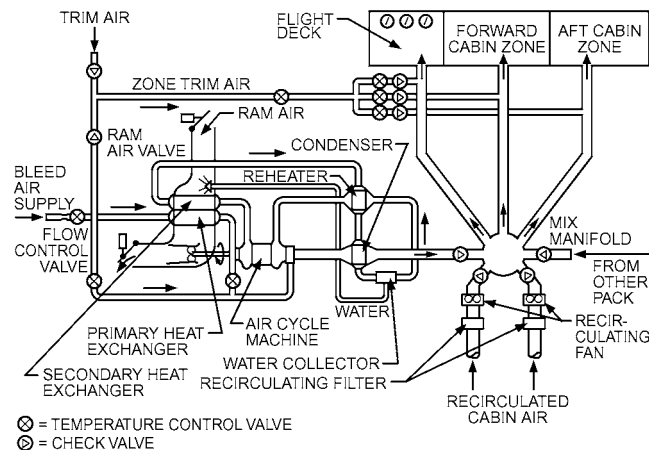


Fig. 12 Aircraft Air-Conditioning Schematic

well area and the tail cone area aft of the rear pressure bulkhead. Other areas include the areas adjacent to the nose wheel and overwing fairing. The temperature control components and recirculating fans are located throughout the distribution system in the pressurized compartments. The electronic pack and zone temperature controllers are located in the electrical/electronic (E/E) bay. The air-conditioning control panel is located in the flight deck. A schematic of a typical air-conditioning system is shown in Figure 12.

Cabin Pressure Control

Cabin pressure is controlled by modulating airflow discharged from the pressurized cabin through one or more cabin outflow valves. The cabin pressure control includes the outflow valves, controller, selector panel, and redundant positive-pressure relief valves. Provisions for negative-pressure relief are incorporated in the relief valves and/or included in the aircraft structure (door). The system controls the cabin ascent and descent rates to acceptable comfort levels, and maintains cabin pressure altitude in accordance with cabin-to-ambient differential pressure schedules. Modern controls usually set landing field altitude, if not available from the flight management system (FMS), and monitor aircraft flight through the FMS and the air data computer (ADC) to minimize cabin pressure altitude and rate of change.

The cabin-pressure-modulating and safety valves (positive-pressure relief valves) are located either on the aircraft skin, in the case of large commercial aircraft, or on the fuselage pressure bulkhead, in the case of commuter, business, and military aircraft. Locating outflow valves on the aircraft skin precludes handling of large airflows in the unpressurized tailcone or nose areas and provides some thrust recovery; however, these double-gate valves are more complex than the butterfly or poppet-type valves used for bulkhead installations. Safety valves are poppet-type valves for either installation. Most commercial aircraft have electronic controllers located in the E/E bay. The cabin pressure selector panel is located in the flight deck.

2. TYPICAL FLIGHT

A typical flight scenario from London's Heathrow Airport to Los Angeles International Airport would be as follows:

While the aircraft is at the gate and the engines have not been started yet, the ECS can be powered by compressed air supplied by the auxiliary power unit (APU), or bleed air from a ground cart. The APU or ground-cart bleed air is ducted directly to the bleed air manifold upstream of the air-conditioning packs. Once started, the

Table 13A Temperature and Relative Humidity Specifications for Collections in Buildings or Special Rooms

Type of Collection and Building	Type of Control	Long-Term Outer Limits ^a	Annual Averages	Seasonal Adjustments from Annual Average ^b	Short-Term Fluctuations plus Space Gradients ^c	Collection Benefits and Risks ^d
Museums, Galleries, Archives and Libraries in modern purpose-built buildings or purpose-built rooms	AA Precision control, no seasonal changes to relative humidity	≥35% rh ≤65% rh ≥10°C ≤25°C	For permanent collections: historic annual average of relative humidity and temperature. In public display areas, human comfort temperatures can apply.	No change to relative humidity Increase by 5 K; Decrease by 5 K	±5% rh, ±2 K	Mold germination and growth, and rapid corrosion avoided. No risk of mechanical damage to most artifacts and paintings. Some metals, glasses, and minerals may degrade if rh exceeds a critical value. Chemically unstable objects deteriorate significantly within decades at 20°C, twice as fast each 5 K higher.
	A1 Precision control, seasonal changes in temperature and relative humidity	≥35% rh ≤65% rh ≥10°C ≤25°C		Increase by 10% rh. Decrease by 10% rh. Increase by 5 K; Decrease by 10 K	±5% rh, ±2 K	Mold germination and growth, and rapid corrosion avoided. No mechanical risk to most artifacts, paintings, photographs, and books; small risk of mechanical damage to high-vulnerability artifact.
	A2 Precision control, seasonal changes in temperature only	≥35% rh ≤65% rh ≥10°C ≤25°C		No change to relative humidity. Increase by 5 K; Decrease by 10 K	±10% rh, ±2 K	(Current knowledge considers the specifications A1 and A2 as causing the same low risk of mechanical damage to vulnerable collections. Slow seasonal adjustment of 10% rh is estimated to cause the same mechanical risk as rapid fluctuations of 5% rh, because of significant stress relaxation occurring within three months of a slow transition.) Chemically unstable objects deteriorate significantly within decades at 20°C, twice as fast each 5 K higher.
	Temperature at or near human comfort					
Museums, galleries, archives, and libraries needing to reduce stress on their building (e.g., historic house museums), depending on climate zone ^e	B Limited control, seasonal changes in relative humidity and large seasonal changes in temperature. ^f	≥30% rh ≤70% rh ≤30°C	For permanent collection: historic annual average of relative humidity and temperature.	Increase by 10% rh Decrease by 10% rh Increase by 10 K Decrease by up to 20 K	±10% rh, ±5 K	Mold germination and growth, and rapid corrosion avoided. Chemical deterioration halts during cool winter periods No risk of mechanical damage to many artifacts and most books. Tiny risk to most paintings, most photographs, some artifacts, some books. Moderate risk to high-vulnerability artifacts. Objects made with flexible paints and plastics that become brittle when cold, such as paintings on canvas, need special care when handling in cold temperatures. Chemically unstable objects deteriorate significantly within decades at 20°C, twice as fast each 5 K higher. Chemical deterioration halts during cool winter periods.
	C Prevent relative humidity extremes (damp or desiccation) and prevent high temperature extremes.	≥25% rh ≤75% rh ≤40°C ^g		Within 25% to 75% rh year-round. Temperature usually below 25°C	Not continually above 65% rh for longer than <i>X</i> days. ^h Temperature rarely over 30°C	Mold germination and growth, and rapid corrosion avoided. Tiny risk of mechanical damage to many artifacts and most books; moderate risk to most paintings, most photographs, some artifacts, some books; high risk to high-vulnerability artifacts Even greater care is needed than provided in B when handling objects made with flexible paints and plastics that become brittle when cold, such as paintings on canvas. Chemically unstable objects deteriorate significantly within decades at 20°C, twice as fast each 5 K higher.
	D Prevent very high relative humidity (dampness)	≤75% rh		Relative humidity reliably below 75% rh	Not continually above 65% rh for longer than <i>X</i> days. ^h	Chemically unstable objects deteriorate significantly within decades at 20°C, and twice as fast each 5 K higher. Conversely, cool winter season can extend their life. Mold germination and growth, and rapid corrosion avoided. High risk of sudden or cumulative mechanical damage to most artifacts and paintings because of low-humidity fracture; but avoids high-humidity delamination and deformations, especially in veneers, paintings, paper, and photographs.

Table 13B Temperature and Relative Humidity Specifications for Collections in Buildings or Special Rooms

Type of Collection and Building	Type of Control	Specifications	Collection Benefits and Risks
Temporary exhibit space and unpacking space for loaned objects	Conditions will be stipulated in loan agreements ⁱ	Conditions will be agreed between lender and borrower. Based on the historic climate to which the object is accustomed, and a risk assessment of the borrower's environment and that of the transit process. Solutions to protect objects from climate shock should first be found in the creation of microclimates (showcases, glazing, etc., potentially using buffering). ^{c,i}	Benefits and risks are assessed by the lender, and contractual specifications based on this assessment. Often, assessment is highly risk averse, precautionary. For the borrowing institution, the benefits are increased access to popular objects by visitors; risks are monetary and reputational damage if climate control does not meet conditions outlined in the loan contract.
Chemically unstable organic materials in modern purpose-built buildings or purpose-built rooms^j	Cool	8 to 16°C, 30 to 50% rh As defined in ISO <i>Standard</i> 18934:2011. IPI (Adelstein 2009) uses an anchor of 12°C.	The benefit of low temperature storage is extended lifetime of objects that will be lost within a generation or two at room temperature. See the section Chemical Damage for details on quantifying the benefits. Biological damage is also much reduced. The risks are the many side-effects of such systems: high humidity or condensation during malfunctions, water exposure. Objects must be packaged appropriately to reduce risk of condensation during retrieval, and a transition space with intermediate climate may be required. Hourly, daily, and even longer humidity fluctuations do not affect most properly packaged objects at low temperatures. ^e
	Cold	0 to 8°C, 30 to 50% rh As defined in ISO-18934:2011. IPI (Adelstein 2009) uses an anchor of 4°C.	
	Frozen	−20 to 0°C, 30 to 50% rh As defined in ISO <i>Standard</i> 18934:2011 and Adelstein (2009)	
Unstable metal or glass in modern purpose-built buildings or purpose-built rooms	Relative humidity controlled to avoid a critical relative humidity of a salt or hydrate	Many different critical relative humidities for various materials. See the section on Critical Relative Humidity for details and sources of information.	

Notes for *Tables 13A and 13B*:

^aLong-term limits apply to combination of selected annual average plus selected seasonal adjustments. See [Figure 15](#) for examples on a psychrometric chart.

^bRate of seasonal adjustments in relative humidity set point should not exceed the short-term fluctuation limit each 7 days (e.g., for A1, a seasonal adjustment can be no faster than 5% rh change per 30 days and 2 K change per 30 days).

^cShort-term fluctuation means any fluctuation shorter than the times specified in footnote b for rate of seasonal adjustment (i.e., 30 days for relative humidity fluctuations, 7 days for temperature fluctuations). Space gradient refers to the differential in relative humidity or temperature between any two locations where objects are permitted to be placed in the controlled space (designers can specify out-of-limit locations, such as a specific distance to exterior walls and supply vents).

^dSee [Table 3](#) for examples of objects in each sensitivity category, and [Table 5](#) for lifetimes of objects at various temperatures.

^eMicroclimates (enclosures, packaging) can achieve the same relative humidity control as type AA or A in a much less controlled space (e.g., B, C, or D), and with much greater long-term reliability. See the section on Response Times of Artifacts.

^fLong-term risk (≥10 years) of mechanical damage because of relative humidity fluctuations is dominated by the probability of extreme events such as system overload or failure in winter. Control type B with high reliability is less risk to collections than AA or A with poor reliability.

^gAn upper temperature limit is provided for a mixed collection that may contain objects with waxy materials that deform irreversibly beginning at ~40°C. This limit is set more cautiously for type B control, 30°C than type C control.

^hFrom [Figure 3](#), mold germination becomes very slow, but not impossible, in the range of 75 to 65% rh.

ⁱIn general, professional guidance currently refers to Bizot, which stipulates outer limits of 40 to 60% rh, and 16 to 25°C throughout the year. Ratified as of 2016 by ICOM-CC, IIC, AIC, AAMD, NMD, BM, and Bizo. See Michalski (2016) for details.

^jSee [Table 5](#).

gradients. Rather than defining a specification and then estimating the benefits and risks, [Tables 13A](#) and [13B](#) consider practical categories of benefit and risk, and then define the range of specifications consistent with those benefits and risks.

Column 3 (long-term outer limits) specifies the boundaries beyond which risk climbs unacceptably for many mixed collections (in broad agreement with recent guidelines such as BSI PAS *Standard* 198:2012). The upper limit of relative humidity is based on mold risk (see the section on Biological Damage). The lower limits of relative humidity and temperature are based on mechanical risk, such as the probability of fracture of organic materials (see the section on Mechanical Deterioration). The upper limit of temperature is based on the risk of chemical decay, which climbs exponentially with increase in temperature (see the section on Chemical Deterioration). These generalized limits for mixed collections do not replace a thorough determination of the specific vulnerabilities of specific collections based on information in the section on Environmental Effects on Collections, alongside consultation with conservators and scientists. For example, a (clean) stone sculpture collection is not at risk from high summer relative humidity or high temperature (pollution and vandalism are more likely risks).

Column 4 (annual averages) assumes design for permanent collections, not loans. To minimize mechanical risk, and to reduce energy costs and building stress, annual averages can be set at local historic annual averages, to which the collection has mechanically acclimatized. In public display areas, a range of human comfort temperatures can apply, but cannot be set beyond the long-term outer limits.

Columns 5 (seasonal adjustments) and 6 (short-term fluctuations) are similar to older versions of this table, although some ranges are now wider. Seasonal adjustments are constrained by the long-term outer limit, although short-term fluctuations are allowed to extend beyond this limit. For a discussion of dual set-point control as a means to achieve these parameters, see the section on Controls Design.

[Figure 15](#) shows the interrelation of the four specification components and the role of long-term outer limits for an example of control type A1. The long-term outer limit (35 to 65% rh, 10 to 25°C is defined by the solid-line box. For this project, the annual average is 21°C and 42% rh, shown by the black dot; seasonal adjustments are ±10% rh, +5 K, and −10 K, although application of these seasonal adjustments is constrained by the upper temperature and lower humidity limits of the long-term outer limit. This combination of annual average and seasonal adjustment is shown by the dashed-line box. Short-term fluctuations of ±5% rh and ±2 K are added, and the

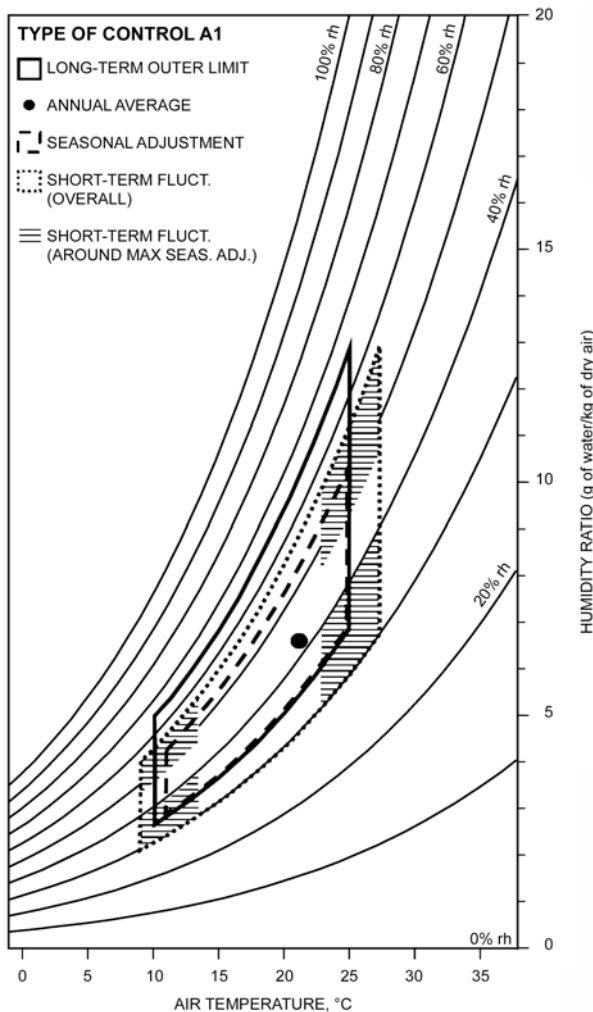


Fig. 15 Psychrometric Depiction of Control Type A1

total range is defined by the dotted-line box. The sections of the dashed-line box that go beyond the long-term outer limits are permissible because they represent short-term fluctuations.

To remain within the bounds of mechanical risk defined for each type of control, it is essential that the annual average be both historically accurate and consistent into the indefinite future. The example in Figure 15, for example, allows short-term relative humidity to drop to 30% because the historic annual average was claimed to be an unusually low 42% rh. At the same time, however, selecting this annual average does not allow short-term humidity to go above 57% rh in summer. Estimates of mechanical risk for each type of control are based on the total range of relative humidity values over many years. If, next year, the annual average setting is changed to 50% rh to justify a summer high of 65% rh, A1 control can no longer be claimed. In a project where historic averages are unknown, select annual averages that are consistent with future needs (e.g., sustainability) or a known critical relative humidity for part of the collection. Given the fixed boundary of the long-term outer limits, the maximum seasonal adjustments are available only for annual averages near the middle of this bounded area.

6.4 AIRBORNE POLLUTANT CONTROL STRATEGIES

In the past, recommendations for maximum pollutant concentrations allowed in museums and archives were based on levels that only

limited numbers of major institutions could achieve, and that were measurable with commercial monitors or with sensitive analytical methods (Mathey et al. 1983; NARA 2002; NRC 1986). In the 1980s, little information existed on the impact of some pollutants, such as acetic acid and nitrogen dioxide. As a precaution, “use best available technology” was the stated advice for those pollutants. This expression became popular and, consequently, many institutions requested it as specification, or requested very low limits of pollutant concentrations without justification. Those low limits were often hard to achieve and maintain. Apart from the cost, it also raised the issue of sustainability. In practice, target levels for pollutants were often simply neglected or ignored.

A common analytic method for measuring specific gaseous pollutants uses **diffusive samplers**. A chemical compound in a diffusion tube absorbs a specific pollutant for a fixed amount of time, typically 3 weeks. After the sampling, collected pollutants are sent to a laboratory for analysis. This method can detect most pollutants of interest for museums with good limits of detection (Grzywacz 2006). Particulate matter of different aerodynamic diameters can be measured with precision using a cascade impactor (Krupinska et al. 2013). However, unlike monitoring of temperature and relative humidity, measuring different pollutants is expensive and many museums will avoid doing it unless there is serious doubt about the actual concentration of some pollutants or damage is reported on an object. Qualitative and semi-quantitative tests include pH testing, which gives an indication of the acidity level (Tétreault 1992); coarse particle deposition on glass or sticky slides (Lloyd et al. 2007); and metal corrosion electronic sensors or metal coupons, which give information on the corrosiveness of the environment (Coughlin 2011; Thickett et al. 2013). Detection limits of some of these tests can be an issue, as can the fact that they may fail to detect the most harmful compounds for the collection. Test results cannot easily be transposed to specific pollutant concentrations. Even with quantitative measurements, monitoring has some limitations. Not all rooms and enclosures are usually tested, and measurements at a specific location and time may provide limited information: pollutant concentrations can vary based on parameters such as changing seasons, crowd density, space gradients, product aging, and HVAC system adjustments. Measuring pollutant concentrations in a new building before its official opening will not give the same results as a building filled with collections, visitors, and older enclosures. It is best to consider a global preventive strategy before starting a monitoring campaign without being sure the results will provide the proper answer.

Table 14 offers a control strategy for pollutants based on a cost benefit scale and on the reduction of uncertainties of the risk evaluation. The table has three levels of control, and makes recommendations based on the building and enclosure; additional considerations are discussed for each level of control.

- **Basic level:** recommended dust filter performance at least equal to that recommended for office spaces (typically minimum efficiency reporting value [MERV] 11) or as specified by an accreditation program such as the U.S. Green Building Council’s (USGBC) LEED Indoor Environmental Quality credit (EQc) 5.1 (MERV 13) (ASHRAE Standard 52.2-2017). If appropriate, enclosures should be well sealed to prevent infiltration of pollutants present in the room. Consultation with conservation professionals can provide information on the global strategy for pollutant control, advise on which objects are typically at risk in museums, and provide guidelines for proper selection of products when building enclosures. The goal of the basic level is to avoid or minimize the most common short- and medium-term damage caused by pollutants in museums and archives, at reasonably low costs.
- **Intermediate level:** dust filtration efficiency should be higher than for the basic level (Tétreault 2003). Qualitative or semiquantitative monitoring is suitable in the new installation (rooms and enclosures), as well as some testing of products before use. Some deeper

Table 14 Strategies for the Control of Airborne Pollutants

Level of Control	Building with HVAC system	Display Cases and Storage Cabinets/Boxes	Considerations
Basic			
Basic control of fine particles and avoiding common problems in enclosures.	- Provide basic fine-particulate filtration such as that recommended for office space regulation or for LEED certification (EQc 5.1). - Locate HVAC fresh air intake away from pollutant sources and keep windows closed.	- In closed spaces containing objects, select and use materials recommended by conservation professionals.^{a, b} - Ensure airtightness of enclosure (to prevent external pollutant infiltration) if there are no significant amounts of pollutants generated by objects or materials (see Table 9).	- Identify objects (e.g., lead, silver, soda-rich glasses, cellulose papers, calcareous objects) that may be at high or moderate risk from pollutants (see Table 9). - Address pollutants by using a systematic approach: avoid, block, dilute, and sorb.^c
Intermediate			
Improved control of fine particles and reduced uncertainty and risk of damage in enclosures.	- Use medium-efficiency fine-particulate filtration or select filter performance based on outdoor concentration provided by local authority. - Seal concrete and wooden surfaces (walls, floor, shelves, etc.).	- Test or investigate materials and objects to identify those that contain harmful compounds.^{b, c} - Monitor enclosed environment with low-cost monitoring techniques (risk of low sensitivity).^{c, d}	- Consider adjusting relative humidity and temperature levels, which often affect pollutants' reactions on objects.
Advanced and special cases			
Optimal control of airborne pollutants in room; better quantification of preservation performance, which allows optimal strategies for improvement.	- Use high-efficiency fine-particulate filtration or select filter performance based on risk analysis result. - Use gas-phase filtration media if outdoor pollutants in surrounding environment or indoor-generated pollutants are an issue.	- Estimate or measure airtightness of enclosure.^c - Options for special needs: positive air pressure,^c gas sorbent,^c anoxia system.^f	- Maximal average pollutant concentrations for a general collection (excluding moderate- and high-risk objects) should stay below 1000 µg/m³ (400 ppb) for acetic acid; 1 µg/m³ (0.7 ppb) for hydrogen sulfide; and 10 µg/m³ for nitrogen dioxide (5 ppb), ozone (5 ppb), and fine particles. These limits should prevent low-level damage to objects for at least 1 year.^c Controlling these key pollutants makes it very likely that other pollutants will be controlled as well. - Quantitatively monitor^d concentration of key pollutants and compare against suggested limits or with institutional targets. - Do risk analysis of outdoor, room, and enclosure pollutant concentrations and determine most efficient solutions for minimizing impact of pollutants on specific objects or on collection in general. Adjust institutional target if necessary.

^aTétreault (2017), ^bHatchfield (2002), ^cTétreault (2003), ^dGrzywacz (2006), ^eCalver et al. (2005), ^fMaekawa (1998).

investigation can be done to identify vulnerable objects and to determine whether emissions from the collections themselves can be a risk to other objects. This will not necessarily improve conservation of the collection from pollutants, but it reduces uncertainties related to the conservation strategies in place. The strategy can be adjusted, if needed, in the light of the results.

- **Advanced level:** quantitative measurements should be taken of the airborne pollutants (gases and fine particles) outside the institution as well as in some rooms and enclosures containing very significant and vulnerable objects. This can be done for a new installation, during renovations, or as needed. The maximum pollutant concentrations allowed can be based either on the limits in Table 14 for a general collection, or on the target for the general collection and/or for some objects established by the institution. Conservation professionals can help assign pollutant target concentrations aligned with the institution's preservation policy. Quantitative measurement of the air exchange rate for enclosures that need a high airtightness is also recommended. Knowing the airtightness also helps determine the quantity of silica gel or any sorbent needed for an optimal climate control in the enclosure.

Measuring particle and gaseous pollutant concentrations and airtightness of enclosures can provide better confidence on the strategy in place, and can support a proper risk analysis for the overall collection or for specific objects (Krupinska et al. 2013). Local environmental data, obtained from different levels of government agencies, can provide useful information on the outdoor climate. This analysis can help determine the filtration performance needed for rooms and for enclosures holding specific objects or collections. If the room is well controlled, leakage from enclosures may not be an issue. How-

ever, if it is difficult to achieve adequate control in the room, then the collection can be better protected inside enclosures. Unfortunately, not all objects can be placed in enclosures (e.g., because of size or access). The length of exhibition/exposure allowed can also be adjusted based on the results of the risk analysis.

For very vulnerable or/and significant objects, some special features can be considered for optimal preservation: positive-air-pressure enclosures (preventing dust infiltration in leaky cases), enclosures with gas sorbents (to reduce the amount of undesired gases generated inside or infiltrated), and low-oxygen enclosures (to minimize oxidative reactions, including photo oxidation). See Table 14 for references.

6.5 CONTROL STRATEGIES FOR OBJECTS WITH HIGH VULNERABILITY TO POLLUTANTS

Some objects tend to be more vulnerable to inadequately controlled environments. Those objects need special considerations that HVAC professionals should be aware of. A conservation professional can also assist with developing preservation strategies.

Silver

Silver is very sensitive to reduced sulfide compounds, mainly hydrogen sulfide (H₂S) and, to some extent, carbonyl sulfide (COS). Sulfur sources are many: the outdoors, from people in the room, and from products and collections inside enclosures. It is usually best to keep silver objects in airtight enclosures with no sulfur-emitting products. Consult the combustion section in the safety material sheet (SMS) for specific products to see if they contain sulfur

compounds; products that contain sulfur compounds should be avoided. It is also wise to confirm the absence of sulfur compounds in the product by running a spot test, such as the lead acetate test, Oddy test, or equivalent (Robinet and Thickett 2003; T  treault 2003). The same strategy can be applied for the preservation of copper. The LOAED for H₂S for silver is 0.10 µg/(m³·yr) (0.071 ppb/yr) and 1.0 µg/(m³·yr) (0.71 ppb/yr) for copper (T  treault 2003). Complete dryness will not stop tarnishing, but will minimize it.

Lead

The most harmful vapor to lead is acetic acid. Lead is not usually at risk of corrosion in a room but may be in enclosures. Any organic-acid-emitting products or objects should be avoided. Lead may never be safe in the presence of wood, painted wood products, or freshly applied sealants or adhesives. The worst situation would be having lead present in a freshly painted enclosure with paint formed by oxidative polymerization (e.g., oil based paint). Polymerization releases aldehydes, organic acids, and peroxides. Those peroxides can convert aldehydes into organic acids (Raychaudhuri and Brimblecombe 2000; T  treault 2011). Enclosing lead objects in a display case freshly sealed with acetoxycured silicone also puts the lead at high risk of corrosion. A relative humidity kept below 35% prevents corrosion by organic acids above the NOAEL (170 ppb).

Calcareous Objects

Calcareous objects (e.g., limestone, ceramics, shells) can react with organic acid vapors, especially when contaminated by chloride or nitrate salts (Halsberghe et al. 2005) in highly humid environments. No data exist to quantitatively assess these objects' vulnerability. As a precaution, it is best to minimize the presence of acid-emitting products or objects in the enclosure as well as relative humidity and temperature fluctuations, and if possible, lower the relative humidity to prevent salt dissolution, reaction, and migration.

Sodium- and Potassium-Rich Glasses

Some historical glasses degrade slowly in the presence of water vapor, resulting in alkali leaching, which can form crystalline corrosion compounds on the surface or modify the structure of the glass. The presence of formic and acetic acids accelerate the leaching (Robinet 2006). These types of glass should be displayed or stored near 40% rh with very little fluctuation (Koob 2006). See van Giffen et al. (2018) for detailed recommendations on climate control for glass. **Enclosures should not contain products that can emit organic acids.**

Colorants

Many colorants (organic pigments and dyes) are known to be sensitive to photooxidation and/or to hydrolysis (Reilly 1998). In addition, some colorants are affected by gaseous pollutants. The most sensitive colorants to nitrogen dioxide, sulfur dioxide, and ozone are curcumin, dragon's blood, aigani, realgar, iron ink, enju, basic fuschin, Brilliant green, pararosaniline, indigo, madder lake, Persian lake, and saffron (Cass et al. 1989; Whitmore and Cass 1989; Williams et al. 1993). Yellow dyes from photographic prints have been found to be affected by acetic acid (Fenech et al. 2010). Artworks with vulnerable colorants should not be displayed long term without protective enclosures, and photograph prints should not be enclosed with products that may release organic acids.

Cellulose Papers

For many decades, sulfur dioxide was thought to be the most damaging pollutant for paper. As its concentration in the environment decreased over the years, it was found that nitrogen dioxide was the main problem for paper. Fine particles and ozone also affect unprotected paper (Bartl et al. 2015; Gurnagul and Zou 1994). At the room level, displaying art on paper without protection (e.g.,

glazed framing, display cases) is not recommended. However, formic and acetic acid emitted by various organic materials can affect cellulose, but in the presence of aldehydes, the damage is found to be reduced (T  treault et al. 2013). As a precaution, however, avoid acid-emitting products.

For paper in books, most damage (yellowing, embrittlement) by outdoor and indoor pollutants tends to remain on the margins of the paper sheets, with very slow diffusion into the book. Many archivists will accept some limited deterioration of the pages' edges. If stack or single-sheet papers are framed or protected in airtight boxes, gas filtration in archives and libraries may not be required. The cellulose is best preserved against acid-catalyzed hydrolysis by keeping the relative humidity and temperature as low as possible.

Cellulose Acetate Films

Cellulose acetate films degrade by acid-catalyzed hydrolysis, and acetic acid is the by-product released (Reilly 1993). It is best to preserve films from the 1950s and 1960s in cool or cold rooms (see Table 13B). In ambient conditions, degraded films should ideally be stored in special ventilated cabinets to avoid the risk of damage to other collections. Otherwise, consider enclosing the films in airtight enclosures with moisture sorbents to prevent the ingress of high humidity in the storage area (Nishimura 2015).

Cellulose Nitrate Films

As with any cellulosic material, cellulose nitrate (CN) films degrade by acid-catalyzed hydrolysis, releasing nitrogen oxides. Old CN films, produced mainly from 1896 to 1952, are unstable and must be kept absolutely below 38  C, above which there is a high risk of self-ignition. CN films should be removed from the collection and properly stored according to NFPA *Standard* 40, which provides detailed information on the ventilation requirements. However, it is best to preserve these films in cold rooms (see Table 13B).

Other CN objects (such as faux tortoise shell) do not degrade to the same magnitude as films, but to avoid the risk of damage from nitrogen oxide emissions to other collections, CN objects should be stored either in well-ventilated rooms or in special ventilated cabinets (Coughlin and Seeger 2008). A room with a high load of CN items must also comply with local regulations for explosive and combustible substances.

Difficult-to-Clean Objects

All objects are susceptible to particle deposition, but cleaning of particles is difficult or even impossible for some objects. During handling and cleaning, there is also a risk of physical damage. Example objects include those with powdery pigments or surfaces (e.g., some painted ethnographic objects, butterfly wings); physically fragile objects (e.g., insect collections, filamentous mineral specimens); objects in which fine particles could become lodged in microcracks or interstices (e.g., ivories, painted objects with cracks); and objects with sticky surfaces (e.g., some deteriorated plastics, some polyethylene-glycol-treated wooden waterlogged objects). For these objects, it is best to display and store them in airtight enclosures or in cases with a positive-pressure system. If enclosure is not an option, it is recommended to maintain a minimum distance between visitors and fragile objects: for example, a distance of 1.5 to 2 m reduces dust deposition by 50 to 75% (Lloyd et al. 2007). This distance prevents deposition of coarse particles on objects, but has limited effect on fine particles because of their longer suspension time.

7. CONTROLS DESIGN

Although control technologies for mechanical systems in cultural heritage institutions are similar to those used in the rest of the HVAC field, the control philosophies and logic that determine daily

operation of systems that condition collections areas can, and typically should, be quite different. A common criticism of collections environments is the amount of energy required to maintain preservation standards and narrow environmental requirements, especially in structures (historic or otherwise) not designed for that level of control. Updated standards (see the section on Key Considerations) paired with an increased understanding of heritage risk (see the section on Overview of Risks) and how collections materials respond to changes in air conditions (see the section on Environmental Effects on Collections) allow for improved approaches to control and operation (see the section on Design Parameters for Performance Target Specifications) that better achieve sustainability goals while providing appropriate preservation conditions for a variety of materials. Applicable standards as well as preservation and sustainability expectations will have been identified during predesign (see the section on Context and Predesign); outcomes of this process may include selecting a nonmechanical solution to manage the collection environment. If a mechanical solution is required or selected, the control and mechanical design must apply the predesign outcomes, including any design parameters for preservation and sustainability. As mechanical systems move from design and construction into a commissioning/continuous commissioning phase, control and operation should be revisited to assess both achievement of the appropriate environmental conditions as well as energy consumption at individual stages of operation; this combination of appropriate environmental preservation conditions while only using the minimum energy consumption necessary is key to long-term optimal, sustainable operation.

In any cultural heritage application, moisture management and control are almost universally the most critical, difficult, and potentially costly processes to achieve. Moisture's role in determining the overall psychrometric properties of any environment, as well as its central role in most forms of collection degradation, make dehumidification and humidification control primary aspects of the holistic building and system operation. Temperature control, though important for both comfort and preservation, is generally the easier control process, and must be managed to maintain appropriate relative humidity at a given moisture content. To facilitate communication with collections professionals, designers and technicians should be prepared to discuss moisture control in the terms with which the client is most familiar and that map well with collections preservation metrics. Relative humidity is typically the best variable for analyzing risk from deterioration processes that depend on sorbed moisture in objects (see the section on Environmental Effects on Collections). Dew point can also be a useful representation of moisture content, especially for discussions of building envelope performance and deterioration (e.g., window and wall condensation), risks during retrieval of objects from cold storage, or entry of loaned objects during winter. Humidity ratio and enthalpy should be clearly defined when used in communication among the broader design team.

Any controls design should clearly define both the control ranges for temperature and relative humidity and the logical process that governs the operation of the relevant equipment. The sequence of operation should be available as a plain-language document that serves as a master reference for institutional staff (collections and facilities) and guidance to outside contractors (designers, programmers, etc.) for controls or equipment upgrades. This master document should be updated as optimization or other changes in operation dictate.

7.1 PHILOSOPHY

As noted previously, environmental tolerances for cultural heritage collections have largely been redefined through updated science, field observations of environmental impacts on collections, and greater awareness of sustainability considerations. Most collections environments can operate safely within a broader range of tempera-

ture and relative humidity conditions than previously understood, leading to new methods and approaches for equipment control and operation to achieve preservation and sustainability goals. Certain environments (e.g., exhibitions with loaned materials governed by an agreement) may still require a narrow band of control, but many collections environments can safely include seasonal adjustments and allow short-term fluctuations without causing damage.

The result is a more complex discussion from the controls design perspective. Information shared and developed during predesign should form the basis of the control philosophy, which should be formalized in a written sequence of operation that provides the logical relationships for how equipment achieves the intended operation. It is critical to recognize that the design sequence of operation is only a model of what is expected to happen: it is likely, even preferable, that the sequence of control will be adjusted during commissioning and optimization. Actual energy loads in the space may be different from models, and it is difficult to predict the effects of collections materials, which may effectively function as heat and moisture sinks in the room environment. Where ratios of hygroscopic material volume to air volume are significant enough, collections may actually buffer environmental changes. Kupczak et al. (2018a) show that, at high ratios, paper collections can reduce fluctuations and energy costs.

Design of temperature and relative humidity controls in cultural heritage settings have traditionally identified a single set point with a dead-band range, and the system works to achieve those set points year-round. With expanded humidity ranges and the use of seasonal temperature control in collections environments, single-point control is no longer the most efficient method. For temperature, occupied collection spaces (where human comfort needs may dictate a narrow range of temperatures throughout the year) are still appropriate candidates for single set-point control. However, low-occupancy collections spaces (e.g., storage) may see both preservation and energy benefits from seasonal temperature adjustments; designs should consider using dual heating/cooling set points or minimum/maximum conditions for seasonal control. Any collections environment, occupied or unoccupied, may benefit from dual relative humidity set points defining where humidification and dehumidification are enabled.

Controls design has two goals:

- Achieving appropriate equipment operation and process management to create and maintain the collection preservation environment defined during predesign
- Using only as much energy as necessary to achieve the desired conditions

Though many cultural heritage institutions or buildings may appear similar on the surface, individual factors such as the following usually require highly individualized controls and equipment designs to match the unique situation (see the section on Context and Predesign):

- Collection type and preservation needs
- Outdoor climate
- Building envelope performance
- Degree of mechanical intervention intended
- Occupancy of collections spaces
- Prioritization of preservation and energy usage
- Institutional capital, operational, and utilities budgets

These factors also heavily influence optimization: similarly constructed buildings and preservation environments (e.g., many off-site library/museum storage facilities) commonly optimize differently for preservation and energy based solely on geographic location and exterior environments, even when many other variables are the same.

drawings and specifications. Both mechanical and electrical specifications must be checked for compatibility and uniformity.

Sequences of Operation

A BAS requires that the engineer define how the system is to be controlled. These sequences are then programmed into the system by the system installer. Writing clear, unambiguous, concise, yet comprehensive sequences of controls is key to the energy and comfort performance of the HVAC system, yet it is very difficult to do well. It requires a clear understanding of how controls work, the limitations of the specific DDC hardware specified and the HVAC system design, and a knack for clear thinking and writing. Techniques for writing successful control sequences are discussed in ASHRAE *Guideline* 13. Sequences for typical HVAC systems can be found in the ASHRAE (2005) CD-ROM, Sequences of Operation for Common HVAC Systems. ASHRAE research project RP-1455 (Hydeman, in progress) investigates sequences of operation for variable-air-volume (VAV) air handlers and terminal units. One goal of the project is to specify control sequences that can be readily implemented by design engineers and controls contractors. Other ASHRAE research projects that may be helpful in developing control sequences include RP-1515, on saving energy by reducing minimum airflow from overhead diffusers (Arens et al. 2014) and RP-1547 (Lau et al. 2013), which addresses demand-controlled ventilation based on CO₂ in multizone systems.

Energy-Efficient Controls

The U.S. Green Building Council's (USGBC) Leadership in Energy Efficient Design (LEED) program for new construction and existing buildings provides guidance on designing and building commercial, institutional, and government buildings to produce quantifiable benefits for occupants, owners, and the environment.

During design and construction, LEED provides many opportunities for using a BAS. LEED awards points, based on design criteria, to determine at which level a building is certified: platinum, gold, silver, or certified. (For specifics, see <http://www.usgbc.org>.) Many of these points depend on the sophistication, flexibility, and power of the BAS, not only for comfort and energy efficiency, but also for sustainability verification throughout the facility's lifetime.

Applications. It is recommended that the LEED design team choose the controls consultant/contractor and to have them involved with the team early in design, allowing for controls input on the control schemes.

For any facility, the BAS provides thermal comfort and routine programming (e.g., occupied/unoccupied). Under LEED, the controls consultant or contractor uses the BAS for benchmarking and alarming when reference points are exceeded (e.g., monitoring facility electrical use). Once a baseline is established, programs with established alarm limits can alert the owner if the facility exceeds the baseline by 10% or more, so corrections can be made and continued higher utility expenses avoided. Depending on the facility, this may be monitored for hourly, daily, weekly, or monthly comparison.

The BAS can also

- Turn solar panels for optimum sun exposure
- Adjust blinds and awnings
- Keep indoor air quality (IAQ) acceptable by adjusting outdoor air (OA) without overventilating
- Open and close windows for natural ventilation
- Monitor and adjust indoor and outdoor lighting
- Control irrigation based on weather
- Collect weather data
- Track scheduling hours for occupancy
- Monitor equipment run times and set points
- Track electricity use profiles

- Monitor efficiency of large equipment performance (e.g., chillers, variable-frequency-drive [VFD] pumping, boilers)

All of these factors affect building sustainability.

Because the BAS operates in real time, and can compare real-time data to previously defined baselines, it is a valuable tool. Open protocols allow the BAS to monitor, control, and provide critical alarming for non-HVAC equipment (e.g., for power monitoring, chiller performance, VFDs for VAV systems or variable-speed pumping, water efficiency, emergency generators, indoor and outdoor lighting, boilers) for a minimal investment.

The BAS provides an excellent tool for commissioning both HVAC and other equipment, as required by LEED. Its data acquisition abilities allow comparisons of daily performance as building use changes, thereby allowing the commissioning agent to determine whether equipment is operating properly, and allowing the owner to compare real-time data to previously defined benchmarks in **measurement and verification (M&V)**. Typical operation sequences, such as optimizing outdoor air quantities or chiller efficiency, and providing multiple sensing points for thermal comfort, are still used, and are critical for maintaining sustainability.

Perhaps the most important aspect of a successful control system is training the owner. Many owners do not fully realize the capabilities of BAS, many of which are intangible and somewhat obscure, so it is critical that the owner be given proper training in using the controls system. The order in which training takes place is equally important: mechanical equipment training should come first, then operation and maintenance (O&M) and controls layout configurations, use of the controls system to provide thermal comfort and maintain equipment, and, finally, written M&V procedures to maintain sustainability.

If the facility cannot justify a full-time energy manager, then the owner should consider third-party contracting to ensure the facility performs at its designed energy efficiency.

7. CONTROL PRINCIPLES FOR ENERGY CONSERVATION

After a building's general needs have been established and the building and system subdivision has been made, the mechanical system and its control approach can be considered. Designing systems that conserve energy requires knowledge of (1) the building, (2) its operating schedule, (3) the systems to be installed, and (4) ASHRAE *Standard* 90.1. The principles or approaches that conserve energy are as follows:

- *Run equipment only when needed.* Schedule HVAC unit operation for occupied periods. Run heat at night only to maintain internal temperature at around 13 and 16°C to prevent freezing. Start morning warm-up as late as possible to achieve design internal temperature by occupancy time, considering residual space temperature, outdoor temperature, and equipment capacity (optimum start control). Under most conditions, heating and cooling equipment can be shut down some time before the end of occupancy, depending on internal and external load and space temperature (optimum stop control). Calculate shutdown time so that space temperature does not drift out of the selected comfort zone before occupancy ends and ensure that ventilation is provided throughout occupancy.
- *Sequence heating and cooling.* Do not supply heating and cooling simultaneously unless it is required for humidity control. Central fan systems should use cool outdoor air, if available, in sequence between heating and cooling. Zoning and system selection should eliminate, or at least minimize, simultaneous heating and cooling. Also, humidification and dehumidification should not take place concurrently.

- *Provide only the heating or cooling actually needed.* Reset the supply temperature of hot and cold air (or water). In air systems that support variable-speed fans, reset duct static pressure to provide thermal comfort at the lowest possible fan speed and energy consumption.
- *Supply heating and cooling from the most efficient source.* Use free or low-cost energy sources first, then higher-cost sources as necessary.
- *Apply outdoor air control.* When on minimum outdoor air, supply no less than that recommended by ASHRAE *Standard* 62.1 and, on VAV systems, include controls that ensure that outdoor air rates are maintained under all expected supply air operating conditions.

System Selection

The mechanical system significantly affects the control of zones and subsystems. System type and number and location of zones influence the amount of simultaneous heating and cooling that occurs. For perimeter areas, heating and cooling should be controlled in sequence to minimize simultaneous heating and cooling. In general, this sequencing must be accomplished by the control system because only a few mechanical systems (e.g., two-pipe and single-coil) can prevent simultaneous heating and cooling. Systems that require engineered control systems to minimize simultaneous heating and cooling include the following:

- *VAV cooling with zone reheat.* Reduce cooling energy and/or air volume to a minimum before applying reheat.
- *Four-pipe heating and cooling for unitary equipment.* Sequence heating and cooling.
- *Dual-duct systems.* Condition only one duct (either hot or cold) at a time. The other duct should supply a mixture of outdoor and return air.
- *Single-zone heating/cooling.* Sequence heating and cooling.

Some exceptions exist, such as dehumidification with reheat.

Control zones are determined by the location of the thermostat or temperature sensor that sets the requirements for heating and cooling supplied to the space. Typically, control zones are for a room or an open area of a floor.

Energy standards such as ASHRAE *Standard* 90.1 no longer allow constant-volume systems that reheat cold air or that mix heated and cooled air, except in special applications such as hospitals. If used, they should be designed for minimal use of reheat through zoning to match actual dynamic loads and resetting cold and warm air temperatures based on the zone(s) with the greatest demand. Heating and cooling supply zones should be structured to cover areas of similar load. Areas with different exterior exposures should have different supply zones.

Systems that provide changeover switching between heating and cooling prevent simultaneous heating and cooling. Some examples are hot or cold secondary water for fan-coils or single-zone fan systems. They usually require small operational zones, which have low load diversity, to allow changeover from warm to cold water without occupant dissatisfaction.

Systems for building interiors usually require year-round cooling and are somewhat simpler to control than exterior systems. These interior areas normally use all-air systems with a constant supply air temperature, with or without VAV control. Proper control techniques and operational understanding can reduce the energy used to treat these areas. General load characteristics of different parts of a building may lead to selecting different systems for each.

Load Matching

With individual room control, the environment in a space can be controlled more accurately and energy can be conserved if the entire system can be controlled in response to the major factor influencing

the load. Thus, water temperature in a water-heating system, steam temperature or pressure in a steam-heating system, or delivered air temperature in a central fan system can be varied as building load varies. Control of the entire system relieves individual space controls of part of their burden and provides more accurate space control. Also, modifying the basic rate of heating or cooling input in accordance with the entire system load reduces losses in the distribution system.

The system must always satisfy the area with the greatest demand. Individual controls handle demand variations in the area the system serves. The more accurate the system zoning, the greater the control, the smaller the distribution losses, and the more effectively space conditions are maintained by individual controls.

Buildings or zones with a modular arrangement can be designed for subdivision to meet occupant needs. Before subdivision, operating inefficiencies can occur if a zone has more than one thermostat. In an area where one thermostat activates heating while another activates cooling, the terminals should be controlled from a single thermostat until the area is properly subdivided.

Size of Controlled Area

No individually controlled area should exceed about 500 m² because of the difficulty of obtaining good distribution and of finding a representative location for space control increases with zone area. Each individually controlled area must have similar load characteristics throughout. Equitable distribution, provided through competent engineering design, careful equipment sizing, and proper system balancing, is necessary to maintain uniform conditions throughout a controlled area. The control can measure conditions only at its location; it cannot compensate for nonuniform conditions caused by improper distribution or inadequate design. Areas or rooms having dissimilar load characteristics or different conditions to be maintained should be controlled individually. The smaller the controlled area, the better the control and the performance and flexibility.

Location of Space Sensors

Space sensors and controllers must be located where they accurately sense the variables they control and where the condition is representative of the area (zone) they serve. In large open areas having more than one zone, thermostats should be located in the middle of their zones to prevent them from sensing conditions in surrounding zones. Typically, space temperature controllers or sensors are placed in the following locations.

- **Wall-mounted thermostats or sensors** are usually placed on inside walls or columns in the space they serve. Avoid outdoor wall locations. Mount thermostats at generally accessible heights according to the Americans with Disabilities Act (ADA) (USDOJ 1994) (usually 1220 mm) and in locations where they will not be affected by heat from sources such as direct sun rays, wall pipes or ducts, convectors, or direct air currents from diffusers or equipment (e.g., copy machines, coffeemakers, refrigerators). The wall itself should be sealed tightly if it penetrates a pressurized supply air plenum either under the floor or overhead. Air circulation should be ample and unimpeded by furniture or other obstructions, and the thermostat should be protected against mechanical damage. Thermostats in spaces such as corridors, lobbies, or foyers should be used to control those areas only.
- **Return air thermostats** can control floor-mounted unitary conditioners such as induction or fan-coil units and unit ventilators. On induction and fan-coil units, the sensing element is behind the return air grille. On classroom unit ventilators that use up to 100% outdoor air for natural cooling, however, provide a forced-flow sampling chamber for the sensing element, which should be located carefully to avoid radiant effect and to ensure adequate air velocity across the element.

If return air sensing is used with a central fan system, locate the sensing element as near as possible to the space being controlled to eliminate any influence from other spaces and the effect of any heat gain or loss in the duct. Where supply/return light fixtures are used to return air to a ceiling plenum, the return air sensing element can be located in the return air opening. Be sure to offset the set point to compensate for heat from the light fixtures.

- **Diffuser-mounted thermostats** usually have sensing elements mounted on circular or square ceiling supply diffusers and depend on aspiration of room air into the supply airstream. They should be used only on high-aspiration diffusers adjusted for a horizontal air pattern. The diffuser on which the element is mounted should be in the center of the occupied area of the controlled zone.
- **CO₂ sensors** for DCV are usually located in spaces with high occupant densities (e.g., conference rooms, auditoriums, courtrooms). Locating the sensor in return air ducts/plenums that serve multiple spaces measures average concentrations and does not provide information on CO₂ levels in rooms with the highest concentrations. CO₂ sensors should be located in the breathing zone of the occupied space (see ASHRAE [2011] and USGBC [2009]).

Commissioning

Commissioning is the process of ensuring that systems are designed, installed, functionally tested, and capable of being operated and maintained in conformity with the design intent. Commissioning HVAC systems begins with planning and includes design, construction, start-up, acceptance, and training, and can be applied throughout the life of the building.

For HVAC systems, **functional performance testing (FPT)** is an important part of the commissioning process. FPT is the process of determining the ability of HVAC system to deliver heating, ventilating, and air conditioning in accordance with the final design intent. Commissioning is team-oriented and generally involves co-operation of various parties, including the owner, design engineers, and contractors and subcontractors. A commissioning authority (the designated person, company, or agent who implements the overall commissioning process) generally leads the process. Each commissioning process must have a plan that defines the commissioning process and is developed in increasing detail as the project progresses. The most useful tool used to challenge (simulate changes to) systems operation is the control system itself. A BAS provides the added convenience of central execution of test steps, and the ability to record responses. The BAS must be commissioned before it can be used to validate the HVAC systems. Commissioning a BAS is discussed in ASHRAE *Guideline* 11. Commissioning HVAC systems is recommended for construction of new buildings, and should be repeated periodically in existing buildings. See [Chapter 43](#) and ASHRAE *Guidelines* 0 and 11 for more information.

REFERENCES

ASHRAE members can access *ASHRAE Journal* articles and ASHRAE research project final reports at technologyportal.ashrae.org. Articles and reports are also available for purchase by nonmembers in the online ASHRAE Bookstore at www.ashrae.org/bookstore.

- Ahmed, O. 2001. A model-based control for VAV distribution systems. Presented at CLIMA Conference, Napoli.
- Ahmed, O. 2002. *Variable-volume, variable-pressure system*. International Facilities Management Association, Research and Development Council, Orlando, FL.
- ANSI. 2013. Non-recirculating direct gas-fired industrial air heaters. ANSI Standard Z83.4-2013/CSA Standard 3.7-2013. American National Standards Institute, Washington, D.C., and Canadian Standards Association, Toronto.

- Arens, E., H. Zhang, T. Hoyt, S. Kaam, J. Goins, F. Bauman, Y. Zhai, T. Webster, and B. West. 2014. Thermal and air quality acceptability in buildings that reduce energy by reducing minimum airflow from overhead diffusers. ASHRAE Research Project RP-1515, *Final Report*.
- ASHRAE. 2003. *Risk management guidance for health, safety, and environmental security under extraordinary incidents*. Presidential Ad Hoc Committee for Building Health and Safety Under Extraordinary Incidents.
- ASHRAE. 2005. *Sequences of operation for common HVAC systems*.
- ASHRAE. 2011. *Standard 62.1-2010 user's manual*.
- ASHRAE. 2005. The commissioning process. *Guideline* 0-2005.
- ASHRAE. 2007. HVAC&R technical requirements for the commissioning process. *Guideline* 1.1.
- ASHRAE. 2009. Field testing of HVAC controls components. *Guideline* 11.
- ASHRAE. 2007. Specifying direct digital control systems. *Guideline* 13-2007.
- ASHRAE. 2010. Selecting outdoor, return, and relief dampers for air-side economizer systems. ASHRAE *Guideline* 16.
- ASHRAE. 2013. Ventilation for acceptable indoor air quality. ANSI/ASHRAE *Standard* 62.1-2013.
- ASHRAE. 2013. Energy standard for buildings except low-rise residential buildings. ANSI/ASHRAE/IES *Standard* 90.1-2013.
- ASHRAE 2012. BACnet®—A data communication protocol for building automation and control networks. ANSI/ASHRAE *Standard* 135-2012.
- ASHRAE. 2013. Ventilation of health care facilities. ANSI/ASHRAE/ASHE *Standard* 170-2013.
- ASME. 2012. *Controls and safety devices for automatically fired boilers*. American Society of Mechanical Engineers, New York.
- Deng, S.-M., and J. Burnett. 1997. Performance monitoring and measurement for central air conditioning chiller plants in buildings in Hong Kong. *HKIE Transactions* 4(1).
- EDR. 2009. *Advanced variable air volume system design guide*. Energy Design Resources, Pacific Gas and Electric Company.
- Englander, S.L., and L.K. Norford. 1992. Saving fan energy in VAV systems—Part 2: Supply fan control for static pressure minimization. *ASHRAE Transactions* 98(1):19-32.
- Erpelding, B. 2007. Real efficiencies of central plants. *HPAC Engineering* (May).
- Felker, L.G., and T.L. Felker. 2010. *Dampers and airflow control*. ASHRAE.
- FEMA. 2007. Reference manual to mitigate potential terrorist attacks against buildings. *Publication* 426, Federal Emergency Management Agency, U.S. Department of Homeland Security, Washington, D.C.
- FGI. 2018. *Guidelines for design and construction of health care facilities*. Facility Guidelines Institute, Dallas.
- Hartman, T. 1993. Terminal regulated air volume (TRAV) systems. *ASHRAE Transactions* 99(1):791-800, *Paper* CH-93-03-3.
- Hartman, T. 2005. Designing efficient systems with the equal marginal performance principle. *ASHRAE Journal* 47(7):64-70.
- Hartman, T. 2012. New strategies for optimizing building performance. *HPAC Engineering*. hpac.com/building-controls/new-strategies-optimizing-building-performance.
- Hydeman, M. In progress. Advanced control sequences for HVAC systems—Phase 1, air distribution and terminal systems. ASHRAE Research Project RP-1455.
- Kalore, P., O. Ahmed, and M. Cascia. 2003. Dynamic control of a building fluid distribution system. Presented at IEEE Conference on Control Applications, Istanbul.
- Kettler, J. 1995. Minimum ventilation control for VAV systems—Fan tracking vs. workable solutions. *ASHRAE Transactions* 101(2):625-630. *Paper* SD-95-02-3.
- Kettler, J. 2000. Measuring and controlling outdoor airflow. *ASHRAE IAQ Applications* (Winter).
- Krarti, M., C.C. Schroeder, E. Jeanette, and M.J. Brandemuehl. 2000. Experimental analysis of measurement and control techniques of outside air intake rates in VAV systems (RP-980). *ASHRAE Transactions* 106(1):39-52, *Paper* 4369.
- Lau, J., G.K. Yuill, and X. Lin. 2013. CO₂-based demand controlled ventilation for multiple zone HVAC systems. ASHRAE Research Project RP-1547, *Final Report*.
- Nelson, K. 2012. Simulation modeling of central chilled water systems. *ASHRAE Transactions* 118(1), *Paper* CH-12-003.
- NFPA. 2014. National electrical code®. ANSI/NFPA *Standard* 70-2014. National Fire Protection Association, Quincy, MA.

- NIOSH. 2002. Guidance for protecting building environments from airborne chemical, biological, or radiological attacks. DHHS *Publication* 2002-139. National Institute for Occupational Safety and Health, Department of Health and Human Services, Cincinnati.
- Taylor, S. 2002. Primary-only vs. primary-secondary variable flow systems. *ASHRAE Journal* 44(2):25-29.
- Taylor, S. 2007. Increasing efficiency with VAV system static pressure set-point reset. *ASHRAE Journal* 49(6):24-32.
- Taylor, S. 2011. Optimizing design and control of chilled water plants, part 1: Chilled water distribution system selection. *ASHRAE Journal* (July).
- USDOJ. 1994. *ADA standards for accessible design*. Title III Regulations, Appendix A. 28CFR36. U.S. Department of Justice, Washington, D.C.
- USGBC. 2009. *LEED® reference guide for green building design and construction*. U.S. Green Building Council, Washington, D.C.

BIBLIOGRAPHY

- ASHRAE. 2013. Commissioning process for buildings and systems. ANSI/ASHRAE/IES *Standard* 202-2013.
- Avery, G. 1992. The instability of VAV systems. *HPAC Engineering* (February):47-50.
- Federspiel, C.C. 2005. Detecting critical supply duct pressure, *ASHRAE Transactions* 111(1):957-963, *Paper* OR-05-13-1.
- Kettler, J. 2004. Return fans or relief fans: How to choose? *ASHRAE Journal* 46(4):28-32.
- Mills, E., H. Friedman, T. Powell, N. Bourassa, D. Claridge, T. Haasl, and M.A. Piette. 2004. The cost-effectiveness of commercial-buildings commissioning. *Report* LBNL-56637 (rev.). Lawrence Berkeley National Laboratory. PDF available from [sites.google.com/site/evanmillsresearch/home/publications](https://www.google.com/site/evanmillsresearch/home/publications).
- Salsbury, T. and B. Chen. 2002. A new sequence controller for multistage systems of known relative capacities. *International Journal of HVAC&R Research* (now *Science and Technology for the Built Environment*) 8(4): 403-428.
- Seem, J., M. House, G. Kelly, and C. Klaassen. 2000. A damper control system for preventing reverse airflow through exhaust air dampers. *International Journal of HVAC&R Research* (now *Science and Technology for the Built Environment*) 6(2): 135-148.
- Shadpour, F. 2001. *Fundamentals of HVAC direct digital control: Practical applications and design*. Hacienda Blue, Escondido, CA.
- U.S. DOE. 2008. Continuous commissioning process. Ch. 8 in *Commissioning for federal facilities*. U.S. Department of Energy, Washington, D.C. [energy.gov/node/727996](https://www.energy.gov/node/727996).

Corrosion Preventive and Protective Measures

Materials Selection. Any piece of heating or air-conditioning equipment can be made of metals that are virtually corrosionproof under normal and typical operating conditions. However, economics usually dictate common material choices. When selecting construction materials, the following factors should be considered:

- Corrosion resistance of the metal in the operating environment.
- Corrosion products that may be formed and their effects on equipment operation.
- Ease of construction using a particular material.
- Design and fabrication limitations on corrosion potential.
- Economics of construction, operation, and maintenance during the projected life of the equipment. Expenses may be minimized in the long run by paying more for a corrosion-resistant material and avoiding regular maintenance.
- Use of dissimilar metals should be avoided. Where dissimilar materials must be used, insulating gaskets and/or applied coatings must be used to prevent galvanic corrosion.
- Compatibility of chemical additives with materials in the system.

Protective Coatings. The operating environment has a significant role in the selection of protective coatings. The suitability of a coating for a particular environment also depends on the surface preparation of the metal to be protected, coating thickness, and coating application technique. Coating durability depends on the adhesion between the coating and the metal. Adhesion depends greatly on surface preparation, including cleaning, degreasing, and the application of primers.

Defects in a coating are difficult to prevent. These defects can be flaws introduced into the coating during application or mechanical damage sustained after application. To maintain corrosion protection, defects must be repaired both in the fabrication process and in the operational environment. Defects in coatings, called **holidays**, can lead to significant localized corrosion of the base metal, most commonly in the form of oxide lifting or cathodic disbondment. **Oxide lifting** occurs when anodic corrosion products accumulate under a coating adjacent to holidays, lifting the coating from the protected surface. **Cathodic disbondment** is the result of the cathodic reduction of dissolved oxygen, which, along with water, migrates under the coating from holidays. Once under the coating, blistering occurs, separating the coating from the protected surface.

Cycles of Concentration. Some corrosion control may be achieved by optimizing the cycles of concentration (the degree to which soluble mineral solids in the makeup water have increased in the circulating water because of evaporation). Generally, adjustment of the blowdown rate and pH to produce a slightly scale-forming condition (see the section on Scale and Deposit Formation Control) results in an optimum condition between excess corrosion and excess scale.

Chemical Methods. Historically, several chemicals have been used to inhibit corrosion. It is important that the inhibitor not interfere with the heat transfer or other functions of the metal. In general, corrosion inhibitors function in one of three ways:

- **Passivation:** these inhibitors form a thin protective oxide film on the metal surface. Nitrites and molybdates work in this way.
- **Precipitation:** these inhibitors contain materials that are precipitated out of solution to form protective films at sites in a corrosion cell. Zinc and phosphates work this way.
- **Adsorption:** these inhibitors are directly adsorbed onto the metal surface, where they form a protective layer. Organics usually function in this manner.

Many inhibitors work best when initially applied at two to three times their normal dosage for several weeks. This **pretreatment**

procedure improves corrosion control by rapidly establishing the protective inhibitor films to new metal surfaces. This procedure should also be repeated for start-ups that follow shutdowns and any time pH or other deviations occur.

Film-forming chemical inhibitors reduce or stop corrosion by interfering with the corrosion mechanism. Inhibitors usually affect either the anode or the cathode. The most common inhibitors are molybdates, phosphates, zinc, phosphonates, silicates, nitrite-borates, aromatic azoles, and organics. Although very effective, chromate inhibitors are severely restricted because of their environmental toxicity and federal laws that have banned their use. The most important factor in an effective corrosion inhibition program is consistent control of both the corrosion inhibition chemicals and the key water characteristics. No program will be effective without controlling these factors.

Anodic corrosion inhibitors establish a protective film at the anode. They stop the corrosion cell by blocking the electrochemical reaction at the anodic site and reestablishing polarization. Though these inhibitors can be very effective, they can also be very dangerous if insufficient anodic inhibitor is present, because the entire corrosion potential can be focused at unprotected anode sites. Severe localized pitting attack can occur at unprotected anodic sites if insufficient inhibitor is present.

Cathodic corrosion inhibitors retard oxygen reduction at the cathode and slow overall electrochemical activity. Cathodic inhibitors are not as effective as fully deployed anodic inhibitors, but are commonly used in conjunction with anodic inhibitors to reduce anodic inhibitor dosage requirement.

General corrosion inhibitors protect by forming a film on all metal surfaces, whether anodic or cathodic. Optimum corrosion inhibition is typically obtained through use of one or more anodic and/or cathodic inhibitor(s).

Filming inhibitors are chemical additives that form a molecular film on metals to create a barrier to corrosion (analogous to paint but much thinner). A filming inhibitor bonds to metal through direct molecular charge interaction between the inhibitor and the metal's surface due to opposite electrical charge. Such bonding is quite strong. The class of copper corrosion inhibitors termed azoles is a good example.

Cathodic Protection. Sacrificial anodes reduce galvanic attack by providing a metal (such as zinc or magnesium) that is more active (anodic) in the galvanic series than two metals in a galvanic couple. A sacrificial anode becomes anodic to the surface metal and supplies electrons to relatively cathodic surfaces, thereby limiting the corrosion of the more anodic metal in the galvanic couple. Proper design and placement of anodes is important. When properly used, they can reduce loss of steel from the tube sheet of exchangers using copper tubes. Sacrificial anodes have helped supplement chemical programs in many cooling-water and process water systems.

Impressed-current protection is a similar corrosion control technique that reverses the corrosion cell's normal current flow by impressing a stronger current of opposite polarity. Direct current is applied to an anode (inert [platinum, graphite] or expendable [aluminum, cast iron]), reversing the galvanic flow and converting the steel from a corroding anode to a protective cathode. The method is very effective in protecting essential equipment such as elevated water storage tanks, steel tanks, or buried pipelines and structures.

Corrosion Measurement

Uniform, or general, corrosion can be quantitatively measured. Localized corrosion, by definition, may only be observed in some cases, and is unobservable in most cases. Although the process of localized corrosion itself is not measurable, damage to metals by localized corrosion attack can be assessed through various means, such as eddy current testing of chiller tubes.

Corrosion Coupons. Using corrosion coupons is an accepted method for evaluating uniform corrosion rates in both closed- and open-loop systems. Coupon specimens can sometimes exhibit results of localized attack, most frequently from underdeposit and/or microbiologically influenced corrosion. Evidence of excessive localized attack on a coupon specimen may invalidate the coupon specimen from further inclusion in a test or series evaluation. Careful selection, handling, placement, alignment, and recording of coupon specimens is required for accurate test results. Flow rates and velocities of the fluid through the coupon rack are also essential: velocity of 1 m/s is normally used, and velocities in excess of 1.5 m/s may influence erosion, particularly of soft metals such as copper. Ideally, constant flow should be maintained throughout the test period and variable-flow piping systems should use pressure-independent flow control devices in the coupon rack piping. Coupon specimens should not be disturbed during a test.

Mild steel and copper are commonly used as coupons; admiralty grades of brass and various grades of stainless steel are less common. Regardless, coupon selection should reflect system metallurgy. Placement of the least noble coupon specimen should precede progressively more noble coupons in the rack in the direction of fluid flow.

Because of high initial corrosion rates of the specimen coupon metals, long-duration tests of coupons (60 to 90 days) yield more accurate results in operating systems than shorter tests (15 to 30 days). For comparison and trend analysis, a series of tests should be conducted using the same or similar exposure lengths.

Electronic corrosion sensors or probes can be used to determine uniform corrosion rates in a system or to augment a corrosion coupon program. Many different types of probes and systems are available, including electric resistance and inductive resistance probes. Linear polarization resistance probes give real-time feedback of corrosion rates. However, because rates of uniform corrosion are rarely consistent over time, the readings from linear polarization devices should be monitored or logged to assess trends in the corrosion rates and system stability. Electronic sensors do not exhibit the clear differences between uniform and localized attack that may be observable on coupon specimens. Consult a water treatment professional about the proper type of probe for the specific application and water treatment program.

2.3 BIOLOGICAL GROWTH CONTROL

Heating equipment operates above normal biological temperature limits and therefore has fewer microbial problems. Hydronic systems operated below about 50°C provide conditions for growth and proliferation of microorganisms: abundant water, aeration, a possible supply of nutrients, and surfaces upon which to attach. Nutrients may include inorganic and organic compounds added directly to fight corrosion/scale/foam (e.g. nitrites) or introduced as a process or operational contaminant. Problems associated with the uncontrolled growth of microorganisms in cooling or low-temperature heating water systems can be categorized in the following four areas:

- Plugging and fouling by biological slime (biofilm/biomass)
- Biodeterioration (e.g., wood rot)
- Microbiologically influenced/induced corrosion (MIC)
- Potential health/safety issues (e.g., disease-causing bacteria)

Generally speaking, microbiological activity depends on, and reacts to, the specific conditions of a given water system. Clean, municipally treated, drinking (potable) water likely has extremely low biological activity. Wastewater and sewage are most likely highly biologically active. There are a vast number of different biological entities that can potentially be found in water, and the topic of microbiology is far too complex for involved discussion in this

context, but there are a few key microbiological concepts very important to water treatment:

- All practical water systems have some degree of biological activity.
- Carbon is the primary component of biological activity, along with nitrogen and phosphorous.
- Biological activity appears in two forms: freely circulating organisms (planktonic) and organisms attached to surfaces (sessile); the same organism types may be found circulating or in biofilms.
- Sessile organisms form biofilm, which is present in all water conduits to a degree (even human arteries and in the human digestive tract).
- Metabolism is an oxidation/reduction process involving transfer of electrons from an electron source (food/nutrient) to an electron acceptor, most commonly oxygen. Through this process, an organism obtains energy and the building blocks for growth and reproduction.
- Metabolism is fostered by biological catalysts (enzymes).
- Metabolism produces inorganic and organic waste products, which tend to be acidic/corrosive.
- Some organisms (aerobic) require oxygen to maintain metabolism and die or go dormant in the absence of oxygen (**obligate aerobes**).
- Some organisms (anaerobic) only use other electron acceptors (oxidizing agents), such as sulfate, and die or go dormant in the presence of oxygen (**obligate anaerobes**).
- Some organisms can metabolize with or without oxygen (**facultative anaerobes**).
- Chemical/biochemical conditions (e.g., pH, oxygen level, enzyme presence) may be considerably different under biofilm than in general water circulation (bulk phase water).
- Biofilm mass depends on nutrient availability, temperature, and oxygen.
- Small particles of biofilm, containing a large number of cells, freely detach from biofilm to migrate within a system and promote biofilm extension

Biological Categories

Microorganisms found in cooling-water systems generally fall into one of three categories: (1) bacteria, (2) algae, or (3) fungi. These organisms can enter a cooling system in various ways: through the makeup water supply; from the atmosphere during normal operation; or from accumulations of environmental organic matter such as insects, bird droppings, grass clippings, and construction debris. Microbial growth can interfere with a cooling operation by causing fouling or corrosion, and may pose a health hazard.

Bacteria. When considering biological growth in a cooling system, it is important to distinguish between planktonic (free-living) and sessile (attached) microorganisms:

- **Planktonic bacteria** are suspended in the water and sometimes referred to as “free floaters” or “swimmers.” These are aerobic bacteria that thrive in an oxygenated environment. They are not harmful to the cooling system, because they do not directly cause deposits or corrosion, but they can provide nutrients for other microorganisms. Generally speaking, the count of planktonic bacteria increases with increasing organic carbon nutrient as measured by the water’s chemical oxygen demand (COD) or its total organic carbon (TOC) content. In addition, some planktonic bacteria, such as *Legionella pneumophila*, are pathogenic and can present a significant human health risk.
- **Sessile bacteria** are nonswimming (attached) bacteria, sometimes called “stickers.” Sessile organisms cause the majority of the corrosion problems in cooling-water systems, because they are not detected in total bacteria counts and tend to be overlooked until problems arise. They establish themselves by attaching to contact surfaces to develop microcolonies with dense biomatrix

structures. Sessile bacteria types include slime formers and anaerobic (corrosive) bacteria. Slime formers can grow and form gelatinous deposits on almost any surface in contact with the cooling water. These deposits can grow so large that they restrict water flow and interfere with heat transfer; they also may promote underdeposit corrosion. Usually, if slime formers are present in the system, deposits can be felt on the sides of the cooling tower basin just below the water level. Anaerobic bacteria thrive in oxygen-deprived environments and often establish colonies beneath slime deposits or under other types of deposits. Surface microbial measurements using special coupons and/or other monitoring devices can be used to monitor sessile bacteria.

Biological fouling can be caused by a wide variety of organisms that produce biofilm and slime masses. Slimes can be formed by bacteria, algae, yeasts, or molds and frequently consist of a mixture of these organisms combined with organic and inorganic debris. Among the problematic bacterial species, sulfate-reducing bacteria (SRB) and iron-related bacteria (IRB) are the most concerning species (after *Legionella*). Their presence suggests a potential for severe corrosion and fouling. IRB are commonly found in tubercles and account for assimilation of ferrous ions to form mounds of ferric hydroxide, whereas SRB are usually found in the bottom of tubercles with black iron sulfides and sometimes elemental sulfur. The sulfur products give off a rotten egg smell. They signal the presence of acid-producing bacteria and deep penetrating pits.

Because biofilm microorganisms can dramatically enhance, accelerate, and, in some cases, initiate localized corrosion (pitting), they are tested and monitored for critical heat transfer operations. Monitoring methods include the following:

- Visual and tactile inspection
- Water analysis for ammonia, chemical oxygen demand (COD), total organic carbon (TOC), adenosine triphosphate (ATP), bacteria speciation's, and planktonic plate counts
- Online biofilm monitors, including corrosion-resistant coupons, disks/plugs, electrochemical sensors, thermal gradient sensors, and sidestream devices to track hydrodynamic pressure differential and heat transfer

Microorganisms can influence localized corrosion directly by their metabolism or indirectly by the deposits they form. Corrosion may not be mediated by simply killing the planktonic microorganisms. Biofilm removal through chemical and/or mechanical cleaning is usually necessary. Corrosion can be substantially mediated by inhibiting microorganism metabolism with proper choice of a biocide program.

Algae uses energy from the sun to convert bicarbonate or carbon dioxide into biomass. Algal mass can block piping, distribution holes, and nozzles.

Control Measures

Eliminating sunlight from wetted surfaces such as distribution troughs, cooling media, and sumps significantly reduces algae growth. A distribution deck cover, which drastically reduces the sunlight reaching the algae, is one of the most cost effective control devices for a cooling tower.

Eliminating dead legs and low-flow areas in the piping and the cooling loop reduces biological growth in those areas. Careful selection of materials of construction can remove nutrient sources and environmental niches for growth and also helps mitigate corrosion. Maintaining a high-quality makeup water supply with low bacterial counts reduces biological growth. Equipment should be designed with adequate access for inspection, sampling, and manual cleaning.

Mechanical aids include submicron filter media, fine strainers, scrapers/brushes, and flow and temperature control. Nonchemical

treatment approaches include biocidal paint, ultraviolet light, sonication with ultrasound, and modification of contact surface material.

Sometimes, effective control of slime and algae requires a combination of mechanical and chemical treatments. For example, when a system already contains a considerable accumulation of slime, a preliminary mechanical cleaning makes the subsequent application of a biocidal chemical more effective in killing the growth and more effective in preventing further growth. A build-up of scale deposits, corrosion products, and sediment in a cooling system also reduces the effectiveness of chemical biocides. Routine disinfection of cooling towers, including the use of high-level chlorination and a biocidispersant, helps control *Legionella* as well as other microorganisms. Sidestream filtration and monitoring devices such as corrosion coupons, test spools, and biofilm sensors can enhance control of biofilm growth. Alternating two different types of biocide can overcome resistance developed by bacteria against a single biocide. The two different types should include two different kill mechanisms.

More cooling-water treatment programs fail because of lack of microbiological control than from any other treatment problem. Scale, corrosion, and fouling are often symptoms of poor microbiological control. Corrosion often occurs under bacterial slime layers. Inorganic foulants are trapped in slime layers, compounding problems. The effects of uncontrolled microbiological activity and fouling may negate the effectiveness of even the best programs for scale prevention and corrosion inhibition. Appropriate biocides must be selected to handle the most prevalent problems first, perhaps in conjunction with dispersants to penetrate and remove deposits. Effective microbiological control is an absolute necessity for a open cooling-water program to be successful.

An effective microbicide (biocide) program has four steps:

- Identify the types and concentrations of microorganisms present in the system.
- Select proper biocides based on system design, discharge restrictions, and types of microorganisms.
- Implement proper application, dosage, contact time, and control of the selected biocides.
- Perform seasonal disinfection cleanings.

Microbicides. Chemical biocides used to control biological growth in cooling systems fall into two broad categories: oxidizing and nonoxidizing. There are numerous options commercially available. Some address a broad spectrum of organisms, whereas others are used to target specific types of organisms such as algae, SRBs, or fungi. Selection of the biocides should always be left to water treatment professionals.

Effective biocide treatment is usually a combination of an oxidizing biocide with a nonoxidizing biocide. The oxidizer may be fed continuously or intermittently. Nonoxidizing biocide may be introduced with either oxidizing biocide strategy. Oxidizing biocides provide quick-kill results and are best used to control biofilm; nonoxidizers are relatively slow acting and best used to suppress MIC activity. A biocidispersant (surfactant) is often also incorporated to enhance biocide performance.

Oxidizing Biocides. These substances (chlorine, bromine, ozone, iodine, chlorine dioxide, and halogen-releasing compounds such as bromochlorodimethylhydantoin [BCDMH or BCD]) are among the most effective biocide chemicals used in water treatment. However, they are not always effective or appropriate for use in cooling-water systems with high organic loading, high temperatures, and/or systems with corrosion or contaminant (compatibility) issues. Chlorine dioxide is probably the one oxidizing biocide least affected by high chemical oxygen demand (COD) and total organic carbon (TOC). In air washers, the odor may become offensive to building/facility occupants, or the exit air (to the building/facility) may be too corrosive for the environment. In wooden cooling towers, high levels of oxidizing biocides can cause delignification.

Overdosing of oxidizing biocides may cause corrosion issues with metallic components in any system. Accurate feed and control of oxidizing biocides for the most effective, efficient, and safe application requires using appropriate feed and control equipment. In many systems, the most effective use of oxidizing biocides is to maintain a constant, low-level residual in the system. However, if halogen-based oxidizing biocides are fed intermittently (slug dosed), a pH below 8 is most advantageous, because of the pH-dependent dissociation curve where the halogen (particularly chlorine) is more prevalent in the more biocidal active (hypohalous acid) form. The residual oxidizing biocide concentration should be tested on a routine basis. Most halogenation programs can benefit from using dispersants or surfactants to penetrate and break up microbial masses and biofilm.

Chlorine has been the oxidizing biocide of choice for many years, either as chlorine gas, or in liquid form as sodium hypochlorite. Other forms of chlorine are available, such as powders, pellets, or calcium hypochlorite sticks; chlorine-releasing compounds (e.g., chloramines, chloroisocyanurates). It can also be produced electrolytically with brine on site. Use of chlorine gas is declining because of health and safety concerns involved in handling it and because of environmental pressures concerning the formation of EPA-regulated halogenated disinfection by-products. In hot-water systems, including potable hot-water systems, chlorine loses efficacy up to 70% at temperatures of 60 to 71°C.

Bromine is usually produced through reactions of an oxidizer such as sodium hypochlorite with sodium bromide on site, or by release from solids of chlorobromohydrantoin compounds. Bromine has certain advantages over chlorine: it is less volatile, and bromamines break down more rapidly than chloramines in the environment. Also, when slug feeding biocide in high-pH systems, hypobromite ions may have an advantage because their biocidal power is better than that of hypochlorite ions. This effect is less important when biocides are fed continuously at low dosage. Like chlorine, bromine can affect the environment with formation of EPA-regulated halogenated disinfection by-products.

Ozone is a gaseous form of oxygen consisting of three oxygen atoms (O_3). As a biocide, it has several advantages compared to chlorine: it does not produce halogenated disinfection by-products, it breaks down to nontoxic compounds rapidly in the environment, it is more potent than chlorine in destructing biofilms, and it requires significantly less chemical handling. The disadvantages of ozone are its short half-life, instability for storage, and reaction by-products that are more biodegradable than their precursors. Use of ozone-generating equipment in an enclosed space requires safety protocols to protect operators from the toxic gas.

Water conditions should be reviewed to determine the need for scale and corrosion inhibitors and then, as with all oxidizing biocides, inhibitor chemicals should be carefully selected to ensure compatibility. To maximize the biocidal performance of ozone, the injection equipment should be designed to provide adequate contact of the ozone with the circulating water. In larger systems, ensure that the ozone is not depleted before ozonated water has circulated through the entire system.

Iodine is provided in pelletized form, often from a rechargeable cartridge. Iodine is a relatively expensive chemical for use on cooling towers and is probably only suitable for use on smaller systems.

Chlorine dioxide gas is effective in low concentrations (0.2 to 0.5 mg/L residual). It must be generated on site. Unlike ozone, it is not affected by pH, and has a longer half-life and therefore a higher penetrating power because of lower oxidizing potential. It is usually generated with sodium chlorite precursor by either an electrolytic generator or reactions with acids. Its drawbacks are instability in storage, costly generating equipment, and precursors and acid salt in the end-product solution.

Hydrogen peroxide (H_2O_2) is a liquid that is usually available in concentrations of 50% or less, by mass, in water. Hydrogen peroxide is considered one of the most environmentally friendly oxidizing biocides, because it degrades to water; however, concentrated hydrogen peroxide reacts in a violent manner when it comes into contact with organic chemicals and materials.

Nonoxidizing Biocides. These biocides destroy organisms either by poisoning the organism, blocking its ability to uptake energy, or disrupting its protective coating (cellular membranes) and exposing the organism to a hostile environment. Most nonoxidizers are also known as **biostats**, which means they inhibit growth of microorganisms.

When selecting a nonoxidizing biocide, many factors must be considered for its effective and efficient use, including system pH and relevant water chemistry, chemical compatibility with other treatment products and system contaminants, and the system turnover rate, or holding time index (HTI), for adequate microbial contact time. Common nonoxidizing biocides include the following:

- Quaternary ammonium compounds
- Polyquats
- Methylene bis(thiocyanate) (MBT)
- Isothiazolones
- Tetrahydro-3,5-dimethyl-2H-1,3,5-thiadiazine-2-thione (Thione)
- Tributyl tetradecyl phosphonium chloride (TTPC)
- Bis(tributyltin) oxide (TBTO)
- Carbamates/Dithiocarbamates
- 2-(Decylthio)ethanamine (DTEA)
- Glutaraldehyde
- Dodecylguanidine (Guanides)
- Tetrakis(hydroxymethyl)phosphonium sulfate (THPS, TKHPS)
- 2-(tert-butylamino)-4-chloro-6-(ethylamino)-s-triazine (TBZ, Tertbutylazine)
- 2,2-Dibromo-3-nitrilopropionamide (DBNPA)
- 2-Bromo-4-hydroxyacetophenone (BHAP)
- 2-Bromo-2-nitropropane-1,3-diol (Bronopol)
- Bromo-nitrostyrene (BNS)
- Proprietary blends
- Aldehyde release treatments
- Organosulfur compounds

How nonoxidizing biocides are fed is important. Sometimes, the continuous feeding of low dosages is neither effective nor economical. Slug feeding large concentrations to achieve a toxic level of the chemical in the water long enough to kill the organisms present can show better results. Water blowdown rate and biocide hydrolysis (chemical degradation) rate affect the required dosage. The hydrolysis rate of the biocide is affected by the type of biocide, along with the temperature and pH of the system water. Dosage rates are proportional to system volume; dosage concentrations should be sufficient to ensure that the contact time of the biocide is long enough to obtain a high kill rate of microorganisms before the minimum inhibitory concentration of the biocide is reached. The period between nonoxidizing biocide additions should be based on the system half-life, with sequential additions timed to prevent regrowth of bacteria in the water. When two nonoxidizing biocides are used, it is important to select the two biocides based on two different kill modes to overcome resistance developed by microbes.

Other Biocides. Ultraviolet (UV) irradiation deactivates microorganisms as the water passes around a quartz tube containing the UV lamp. The intensity of the light and thorough contact with the water are critical in obtaining a satisfactory kill of microorganisms. Suspended solids in the water or deposits on the quartz tube significantly reduce the effectiveness of this treatment method. Therefore, a filter is often installed upstream of the UV lamp to minimize these problems. Because ultraviolet light leaves no residual biocide in the water, sessile organisms and organisms that do not

pass the light source are not affected by the ultraviolet treatment. Ultraviolet irradiation may be effective on humidifiers and air washers where the application of biocidal chemicals is unacceptable and where 100% of the recirculating water passes the lamp. It is less effective where all microorganisms cannot be exposed to the treatment, such as in cooling towers. Ultraviolet lamps require replacement after approximately every 9000 h of operation because, although they will light much longer, the quartz glass slowly becomes opaque to UV light.

Copper (Cu) and silver (Ag) ions can effectively control microbial populations in water systems when the ionization systems that produce them are properly maintained, operated, and applied. Ions are released into the water via electrochemical means to generate 0.4 to 0.8 mg/L of copper and/or 40 to 80 µg/L of silver. The ions assist in the control of bacterial populations. Copper, in particular, effectively controls algae.

Handling Biocides. All biocides must be handled with care to ensure personal safety. In the United States, cooling-water biocides are approved and regulated through the EPA and, by law, must be handled in accordance with labeled instructions. Maintenance staff handling the biocides must read the safety data sheets (SDS) and be provided with all appropriate safety equipment to handle the substance. An automatic feed system is preferred, to minimize and eliminate handling of biocides by maintenance personnel.

***Legionella* and Legionnaires' Disease**

Legionellosis is any disease caused by *Legionella* bacteria. **Legionnaires' disease**, a form of legionellosis, is a potentially fatal, pneumonia-like illness primarily caused by *Legionella pneumophila* serogroup 2, one of the more than fifty known species of *Legionella*.

Legionella are common warm-water microorganisms that are mostly found in surface waters (lakes, ponds, rivers, streams) but can also be found in groundwater sources, including some soils. They tend to grow in biofilm or slime on the surfaces of lakes, rivers, and streams. *Legionella* adapt to conditions in water distribution systems, thereby sometimes escaping municipal water plant disinfection (chlorination). They can, therefore, be found in domestic (potable) water plumbing and associated building water systems such as cooling towers, spas, water fountains, and other water-use systems. Within these systems, *Legionella* can find favorable conditions for growth and amplification, and pose a risk for human disease.

The mere presence of *Legionella* does not necessarily result in disease. Several conditions and factors must occur: primarily, there must be a sufficient quantity and/or virulent form of the bacteria made transmittable and then transmitted to a susceptible host. Transmission occurs when a host inhales tiny water droplets (mists or vapor aerosols) containing *Legionella* from a water-aerosolizing source. Aspiration of water can also occur during normal drinking and swallowing due to choking or gag reflex. Inhalation can bring infectious *Legionella* to the deep, distal parts (alveoli) of the lungs, where they take over and promote the pneumonia of Legionnaires' disease. The dose of *Legionella pneumophila* (or other species of *Legionella*) required to infect humans is not known, and is most probably influenced by host susceptibility.

Legionella grow well and amplify in warm water environments and systems that provide favorable conditions for bacterial growth and the formation of biofilm. The optimum temperature range for growth is 32 to 41°C. The subsequent use of water from such systems harboring *Legionella*, through faucets, shower sprays, humidifying, aerosolizing (misting) devices, or other spray or drift mechanisms of the system may transmit the bacteria to susceptible hosts.

Legionella pneumophila was first identified in 1977, following an outbreak of disease associated with a Philadelphia host hotel for an American Legion bicentennial convention in 1976. Since then, surveillance systems and research studies have been established around the world. In recent years, with improved capabilities and changes in clinical methods of diagnosis, there has been an upsurge in reported cases in many countries. Environmental studies continue to provide information and identify novel sources of infection, leading to regular revisions of guidelines and regulations. As of reports in 2014, an estimated 8000 to 18 000 people are hospitalized with Legionnaires' disease each year in the United States, according to the Centers for Disease Control and Prevention (CDC), and the disease has a 5 to 30% mortality rate (www.cdc.gov/legionella/about/).

Outbreaks of Legionnaires' disease receive media attention. However, this disease more often occurs as single, isolated cases not associated with any recognized outbreak. When outbreaks occur, they are usually in the summer and early autumn, though cases may occur at any time of year.

Water treatment microbial control programs should also include a risk assessment for *Legionella* control and disease prevention. This includes the effectiveness of treatment products and the program to control *Legionella* and the conditions favorable for its growth and amplification, including the accumulation of biofilm (slime/biomass), the presence of amoebae and other protozoans that harbor it, as well as the potential of the water system to produce aerosols and provide routes of aerosol transmission to susceptible human populations.

See ASHRAE *Standard* 188-2015 and ASHRAE *Guideline* 12-2000 for further information.

2.4 NONCHEMICAL AND PHYSICAL WATER TREATMENT METHODS

Nonchemical water treatment has been used for boiler water, cooling water, potable water, and other process applications for decades. There are multiple classifications of such treatment methods and equipment systems. Adherence to manufacturers' recommendations and the proper application of these technologies is critical to achieving successful results. As with traditional methods of water treatment, the ability to achieve successful results should be adequately evaluated for the water treatment system as well as regularly monitored for effective control of scale, corrosion, and biological contamination. Water treatment technologies, published papers, and case histories should be evaluated in light of water supply chemistry, operating environment, duty, and industry standard performance metrics.

Several nonchemical and physical water treatment equipment technologies are classified in the following paragraphs, along with brief summaries of their basic claims and theoretical mechanisms of action.

Magnetic, or fixed-field magnetic, systems are designed to cause scale-forming minerals to precipitate on suspended solids as opposed to heat exchanger surfaces. This mechanism produces non-adherent particles, often classified as the aragonite form of calcium carbonate versus the hard, adherent calcite. The precipitated particles can then be removed by blowdown, mechanical means, or physical flushing.

Electromagnetic systems induce either a constant or variable magnetic flux density (or both), thereby inducing a localized, variable electric field in the bulk water. **Pulsed-power** and **electrodynamic** field systems generate a combination of variable magnetic flux density with a variable electric field at frequencies greater than the incoming line signal. The induced electric fields alter the surface charge on suspended solids and particulate and force precipitation on these surfaces in the bulk water.

Cavitation systems can cause formation and sudden collapse of low-pressure bubbles in a water stream by means of a mechanical force. The collapse of the bubbles imparts a shock wave and resultant heat locally within the water flow. The energy created during the implosion of bubbles promotes precipitation of minerals in the water stream.

Ionization rods create a static electric field that disperses scale-producing foulants, thereby inhibiting precipitation of minerals on equipment surfaces. The dispersion created by the device's electric field also inhibits biological growth.

Electrolysis systems pass an electric current between two electrodes. Precipitation of minerals occurs at the cathode. Biological growth is inhibited by formation of chlorine gas at the anode; the gas combines with water to form hypochlorite.

Ultrasonic systems impart high-frequency acoustic energy in the ultrasonic range to a stream of water. The sound wave produced by ultrasonic devices can cause mechanical damage to bacteria cell walls. Similar to cavitation devices, ultrasonic energy can also produce low-pressure bubbles, which can collapse and damage bacterial cell walls.

Ultraviolet light systems irradiate a water stream with ultraviolet light. Exposing microorganisms to ultraviolet light inhibits their biological growth. The ability to effectively and efficiently control microbial populations throughout a cooling-water system with a point-of-use ultraviolet light system must be evaluated for any given system.

Ozone systems inject ozone created by coronal plasma, by ultraviolet lamps, or by chemical reaction, into the bulk water. When contact is made with microorganisms, ozone can be a very effective disinfectant for their control. However, it is difficult to maintain an effective residual of ozone throughout a cooling-water system because of its high reactivity and fast dissipation. Ozone is very volatile and can be lost from the system as the water passes through the cooling tower fill. Successful application of ozone for biological control requires that sufficient ozone-generating capacity be provided to sustain a level of ozone residual that will control microbial contamination throughout the cooling-water system.

ASHRAE Research Projects

ASHRAE research projects RP-765 (Nasrazadani and Chao 1996) and RP-747 (Gan et al. 1996) found ozone only marginally effective in scale and corrosion inhibition, respectively. Ozone has been shown to have a very limited and unpredictable effect on calcium carbonate scale, and a limited effect to inhibit corrosion of most metals. It provides some reduction of mild steel corrosion, but increases the corrosion rates of copper and copper alloys. It is also known to attack galvanized steel, and increasing the level of ozone in water increases the corrosiveness of the water.

ASHRAE research project RP-1155 (Cho 2002) studied physical water treatment with respect to scale prevention and/or mitigation. For the study, a physical water treatment (PWT) device was defined as a nonchemical method of water treatment for scale prevention or mitigation. Bulk precipitation on colloidal surfaces was proposed as the mechanism of scale prevention. Three different PWT devices (permanent magnets, a solenoid coil device, and a high-voltage electrode) were tested under laboratory conditions. Fouling resistance data obtained in a heat transfer test section supported the benefit of all three devices when configured in optimum conditions in the laboratory-scale test system.

ASHRAE research project RP-1361 (Vidic et al. 2010) studied physical water treatment and microbial control. In this study, a PWT device was defined as a nonchemical method of water treatment for microbial control. Five different PWT devices (magnetic, pulsed electric field, electrostatic, ultrasonic, and hydrodynamic cavitation) were tested in a pilot-scale cooling tower. No statistically significant difference in planktonic and sessile microbial concentrations was

observed in the (test) pilot-scale cooling tower for any of the devices, as compared to an untreated control tower.

Copies of ASHRAE research project final reports can be obtained from technologyportal.ashrae.org, and are free to ASHRAE members.

2.5 BOILER WATER SYSTEMS

Steam and hot-water systems can be classified into three broad groups: (1) steam and hot water for space heating and humidity control in commercial and residential HVAC systems and many industrial plants; (2) steam for industrial process heating and for use as a process reactant in chemical plants, paper mills, etc.; and (3) steam to drive turbines for electric power generation. The American Society of Mechanical Engineers (ASME) classifies boilers into three basic types: fire-tube, water-tube, and electric (ASME 2015). Boilers are also classified according to pressure, materials of construction, size, heat source, or circulation. A general understanding of boiler design and classifications, as well as requirements of the system, is essential to ensure the system receives proper care to maintain reliability. One of the most important classifications of boiler systems is the end use of the steam or hot water. The focus of this section is on steam and hot-water boiler systems for space heating and humidity control in commercial and residential HVAC systems and some industrial plants.

Hot-water heating and low-pressure (103 kPa or less) steam systems with >95% condensate return are similar to closed-loop cooling systems, except with reversed heat-exchange processes. The hot water or steam adds heat to an HVAC system or to an industrial process. The hot-water temperature is maintained either by heat exchange with a steam source or electric elements, or by direct firing in a hot-water-heating boiler.

Steam generators range from small heating boilers to very large and complex systems generating steam for industrial plants and electric utility stations. All steam boilers operate on certain basic principles. Water is heated to produce steam at a desired pressure. The steam does work by heating a building or a commercial/industrial process. To conserve water and energy, waste steam is condensed after use and a portion of the condensate is returned to the boiler as feedwater, along with required fresh makeup water. All parts of this system, including feedwater preparation, the boiler, and the condensate system, require chemical treatment to protect the equipment, maintain boiler efficiency, and prepare steam with the required quality and purity.

Selection of Water Treatment. As discussed in previous sections, many methods are available to prevent or correct water-caused problems. Selection of the proper water treatment method, and the chemicals and equipment necessary to apply that method, depend on many factors. The chemical characteristics of the water, which change with the operation of the equipment, are most important. Other factors contributing to the selection of proper water treatment are

- Economics
- Chemistry control mechanisms
- Dynamics of the operating system
- Design of major components
- Number of operators available
- Training and qualifications of personnel
- Preventive maintenance program
- Control instrumentation and remote monitoring

The two most significant factors in setting up an economically viable and effective cooling water treatment program are the chemical properties of the water supply and cooling capacity (i.e., rated tonnage).